# The SAMPLE Book

SAMPLE Team

2026-01-08 · Version 1.0.0

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# Documentation Framework

This documentation demonstrates a complete publishing workflow from Markdown source files to professional PDF output.

## Overview

The framework supports:

- **Multi-language content**: Parallel English and German versions
- **Structured navigation**: Hierarchical table of contents with PDF bookmarks
- **Rich formatting**: Tables, code blocks, lists, and images
- **Unicode support**: Extensive language and emoji coverage
- **Metadata management**: YAML frontmatter for document properties

## Document structure

The documentation is organised into:

- **Chapters**: Main content sections
- **Examples**: Demonstration files for various features
- **Appendices**: Supplementary reference material

## Technical features

This framework showcases:

- Reproducible PDF generation
- Font management and fallback chains
- Image asset handling (raster and vector)
- Cross-reference management
- Automated list generation (tables, figures, abbreviations)

# Home

Figure 1: SAMPLE Logo

Welcome to this technical documentation framework demonstration.

## About this document

This publication showcases capabilities of modern documentation systems:

- **Multilingual support**: Parallel English and German versions
- **Rich formatting**: Tables, figures, code blocks, and lists
- **Unicode excellence**: 100+ languages, emoji, and complex scripts
- **Professional output**: High-quality PDF generation with proper typography

## Document structure

The content is organized into:

### Core chapters

Main content demonstrating various documentation patterns and structures.

### Examples

Practical demonstrations of:

- Emoji rendering across categories
- Image formats (raster and vector)
- Language samples and scripts

### Appendices

Supplementary material including:

- Technical specifications
- Font coverage analysis
- Reference materials

## Navigation

Use the table of contents (sidebar or PDF bookmarks) to navigate between sections. Each chapter includes:

- Clear heading hierarchy
- Cross-references where relevant
- Practical examples

## Technical foundation

Built with:

- **Markdown**: Source content format
- **YAML frontmatter**: Structured metadata
- **Python pipeline**: Automated build and validation
- **LaTeX/XeLaTeX**: Professional PDF typesetting

# Dedication

To all contributors to the open-source community who generously share their knowledge, code, and time.

To the pioneers of digital typography who made beautiful, accessible documents possible for everyone.

To the readers who seek understanding through well-crafted documentation.

# Preface

Documentation is the bridge between knowledge and understanding. Well-structured documentation empowers readers to grasp complex concepts, reference critical information, and apply learned principles effectively.

## About this documentation

This document serves multiple purposes:

1. **Demonstration**: Showcasing a complete documentation workflow
2. **Reference**: Providing examples of various documentation patterns
3. **Testing**: Validating the publishing pipeline across different scenarios

## Target audience

This documentation is designed for:

- Technical writers seeking workflow examples
- Developers implementing documentation systems
- Content creators exploring publishing options
- Anyone interested in structured document creation

## How to use this documentation

Readers can approach this document in different ways:

- **Sequential reading**: Follow the chapters in order for a comprehensive understanding
- **Reference use**: Navigate directly to specific sections using the table of contents
- **Example study**: Examine the examples section for practical demonstrations

## Acknowledgements

This documentation framework builds upon established best practices from the technical writing community and leverages modern open-source tools for document processing and PDF generation.

# Chapter 1 – Observable patterns

In software development, we repeatedly encounter similar problems for which proven solutions have been established over time. These recurring structures are referred to as design patterns.

## Historical development

The systematic documentation of design patterns began in the 1990s. Inspired by architecture, where Christopher Alexander described patterns for building construction, software developers transferred this idea to programming.

### Early pioneers

The so-called “Gang of Four” (Gamma, Helm, Johnson, Vlissides) published the seminal work “Design Patterns” in 1994, which categorised and described 23 patterns.

### Modern developments

Today, hundreds of documented patterns exist for a wide variety of application areas – from microservices and reactive programming to cloud architectures.

## Categories of patterns

Design patterns can be divided into three main categories:

### Creational patterns

These patterns deal with object creation and attempt to make object instantiation more flexible:

- **Singleton**: Ensures that only one instance of a class exists
- **Factory**: Encapsulates object creation
- **Builder**: Separates the construction of complex objects from their representation

### Structural patterns

Structural patterns describe how classes and objects can be composed into larger structures:

- **Adapter**: Enables collaboration between incompatible interfaces
- **Composite**: Forms tree structures to represent part-whole hierarchies
- **Decorator**: Dynamically extends objects with additional functionality

### Behavioural patterns

These patterns address the interaction between objects and the distribution of responsibilities:

- **Observer**: Defines a dependency between objects so that changes are automatically propagated
- **Strategy**: Encapsulates interchangeable algorithms
- **Command**: Encapsulates requests as objects

## Advantages of using patterns

Using established design patterns offers several advantages:

1. **Common language**: Teams can communicate complex concepts precisely

2. **Proven solutions**: Patterns have been proven in practice and are well documented
3. **Maintainability**: Code becomes more structured and easier to understand
4. **Flexibility**: Changes can often be implemented with less effort

## Limitations and challenges

Despite their advantages, design patterns are not a panacea:

- **Over-engineering**: Not every problem requires a complex pattern
- **Learning curve**: Understanding and correct application require experience
- **Context dependency**: A pattern must fit the specific situation

## Practical application

When deciding on a design pattern, the following questions should be asked:

1. What problem needs to be solved?
2. Is there an established pattern for this problem?
3. Does the complexity of the pattern justify the expected benefit?
4. Does the pattern fit with the existing architecture?

## Summary

Design patterns are a valuable tool in software development. They provide tested solutions for recurring problems and promote a common technical language. However, their sensible application requires experience and judgement to avoid falling into the trap of over-engineering.

# Chapter 2 – Comparative tables

Tables are an indispensable tool for the structured presentation of information. They enable direct comparison of different options, technologies, or concepts at a glance.

## Fundamentals of tabular presentation

A well-designed table follows clear principles:

### Structure and organisation

| Element | Description | Purpose |
|---|---|---|
| Header row | Contains column labels | Orientation for the reader |
| Data rows | Contain the actual information | Comparable presentation |
| Summary | Optional: sums or averages | Aggregated insights |

### Design principles

Effective tables are characterised by the following features:

1. **Clarity**: Unambiguous column and row labels
2. **Consistency**: Uniform formatting within columns
3. **Readability**: Appropriate line spacing and font sizes
4. **Relevance**: Display only necessary information

## Comparison of programming paradigms

A practical example of using comparison tables is the juxtaposition of different programming paradigms:

| Paradigm | Main features | Typical languages | Application areas |
|---|---|---|---|
| Imperative | Step-by-step instructions | C, Pascal, BASIC | Systems programming |
| Object-oriented | Classes and objects | Java, C++, Python | Enterprise applications |
| Functional | Immutable data | Haskell, Erlang, F# | Data processing |
| Declarative | What instead of How | SQL, HTML, Prolog | Database queries |

### Detailed consideration

Each paradigm has its strengths and weaknesses:

**Imperative programming** - Direct control over flow - Efficient at hardware level - Can become confusing with complexity

**Object-oriented programming** - Modular structure - Reusability through inheritance - Can lead to overhead

**Functional programming** - No side effects - Easy to test - Learning curve for switchers

## Technology comparisons

Comparison tables are particularly suitable for technology decisions:

### Web framework comparison

| Framework | Language | Performance | Learning curve | Community |
|---|---|---|---|---|
| Django | Python | Medium | Medium | Very large |
| Flask | Python | High | Low | Large |
| Spring | Java | Medium | High | Very large |
| Express | JavaScript | High | Low | Very large |
| Rails | Ruby | Medium | Medium | Large |

### Evaluation criteria

Various factors play a role in technology selection:

1. **Performance**: Throughput and response times
2. **Developer productivity**: Speed of development
3. **Maintainability**: Long-term maintenance effort
4. **Scalability**: Growth potential
5. **Ecosystem**: Available libraries and tools

## Database comparison

Another common application area is database comparisons:

| Type | Example | Consistency | Scaling | Use case |
|---|---|---|---|---|
| Relational | PostgreSQL | ACID | Vertical | Transactions |
| Document | MongoDB | Eventual | Horizontal | Flexible schemas |
| Key-value | Redis | Eventual | Horizontal | Caching |
| Graph | Neo4j | ACID | Vertical | Relationships |
| Column | Cassandra | Eventual | Horizontal | Time series |

### CAP theorem

For distributed databases, the CAP theorem is relevant:

- **C**onsistency: All nodes see the same data
- **A**vailability: System always responds
- **P**artition tolerance: System functions despite network failures

According to the CAP theorem, only two of the three properties can be guaranteed simultaneously.

## Best practices for tables

When creating comparison tables, the following points should be considered:

### Content aspects

- Select relevant comparison criteria
- Use objective and verifiable data
- Cite sources where necessary
- Ensure data is current

### Visual design

- Zebra pattern for better readability in long tables
- Highlighting of important rows or columns
- Responsive design for different screen sizes
- Sorting and filtering options for interactive tables

## Summary

Comparison tables are a powerful tool for the structured presentation of complex information. They enable quick comparisons and informed decisions. The key to success lies in carefully selecting relevant criteria and presenting them clearly and consistently.

# Epilogue

Documentation is a living artefact. As technologies evolve and understanding deepens, good documentation grows and adapts to serve its readers better.

## The documentation journey

Creating effective documentation is an iterative process:

1. **Initial creation**: Capturing knowledge whilst it's fresh
2. **Review and refinement**: Improving clarity and accuracy
3. **User feedback**: Learning from readers' experiences
4. **Continuous improvement**: Evolving with changing needs

## Looking forward

The principles demonstrated in this documentation – clear structure, comprehensive examples, and attention to technical detail – remain relevant regardless of the specific tools or technologies employed.

## Final thoughts

Good documentation respects the reader's time and intelligence. It provides clear paths to understanding whilst remaining accessible to those encountering the material for the first time.

May your own documentation efforts be similarly rewarding, both in their creation and in the value they provide to your readers.

# Citation & Footnote Examples

This page demonstrates various citation styles and footnote usage in Markdown documents.

## Footnotes

Markdown supports footnotes[1] that appear at the bottom of the page. You can reference the same footnote multiple times[2].

Here's a longer footnote with multiple paragraphs[3].

Inline footnotes are also possible.[4]

### Named vs Numbered Footnotes

You can use descriptive names for footnotes[5] or just numbers[6].

## Citation Styles

### APA Style (7th Edition)

**Books:**

Smith, J. A., & Johnson, M. B. (2023). *Research Methods in Documentation*. Academic Press.

**Journal Articles:**

Brown, L. K., Davis, R. T., & Wilson, S. E. (2024). Advanced typesetting techniques for multilingual documents. *Journal of Technical Communication*, 45(3), 234-256. https://doi.org/10.1234/jtc.2024.01

**Online Sources:**

Unicode Consortium. (2023, September 12). *Unicode Standard 15.1.0*. https://www.unicode.org/versions/Unicode15.1.0/

### IEEE Style

**Journal Article:**

1 L. K. Brown, R. T. Davis, and S. E. Wilson, "Advanced typesetting techniques for multilingual documents," *J. Tech. Commun.*, vol. 45, no. 3, pp. 234-256, 2024, doi: 10.1234/jtc.2024.01.

**Conference Paper:**

[2] J. A. Smith and M. B. Johnson, "Automated documentation pipelines," in *Proc. Int. Conf. Software Engineering*, London, UK, 2023, pp. 123-130.

---

[1] This is a simple footnote with a reference back to the text.

[2] This is a simple footnote with a reference back to the text.

[3] This is a longer footnote with multiple paragraphs.

You can include additional paragraphs by indenting them.

Even code blocks can appear in footnotes:

```
def example():
    return "footnote code"
```

[4] This is an inline footnote.

[5] Descriptive names make footnotes easier to manage in large documents.

They're especially useful when you need to reorganise content.

[6] Footnotes can be numbered sequentially.

**Book:**

[3] A. Martinez, *Modern Documentation Frameworks*, 2nd ed. Boston, MA, USA: Tech Publishers, 2024.

### Chicago Style (Author-Date)

**Books:**

Martinez, Ana. 2024. *Modern Documentation Frameworks*. 2nd ed. Boston: Tech Publishers.

**Journal Articles:**

Brown, Laura K., Robert T. Davis, and Sarah E. Wilson. 2024. "Advanced Typesetting Techniques for Multilingual Documents." *Journal of Technical Communication* 45 (3): 234-256. https://doi.org/10.1234/jtc.2024.01.

### Zenodo Standard (DOI-based)

Zenodo provides persistent identifiers (DOIs) for research data and publications[7].

**Dataset:**

Smith, John A.; Johnson, Mary B. (2023). Sample Documentation Dataset (Version 1.2) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.1234567

**Software:**

Brown, Laura K.; Davis, Robert T. (2024). GitBook Worker: Automated Documentation Pipeline (v1.0.0). Zenodo. https://doi.org/10.5281/zenodo.7654321

**Publication:**

Martinez, Ana; Wilson, Sarah E.; Thompson, James R. (2023). Best Practices for Technical Documentation. *Zenodo Preprints*. https://doi.org/10.5281/zenodo.8901234

### BibTeX Format

For LaTeX/academic documents:

```bibtex
@article{brown2024advanced,
  title={Advanced Typesetting Techniques for Multilingual Documents},
  author={Brown, Laura K and Davis, Robert T and Wilson, Sarah E},
  journal={Journal of Technical Communication},
  volume={45},
  number={3},
  pages={234--256},
  year={2024},
  doi={10.1234/jtc.2024.01}
}

@software{brown2024gitbook,
  author={Brown, Laura K and Davis, Robert T},
  title={GitBook Worker: Automated Documentation Pipeline},
  version={1.0.0},
  year={2024},
  publisher={Zenodo},
```

[7] Zenodo is an open-access repository operated by CERN, providing DOIs for research outputs including data, software, publications, and more. See https://zenodo.org for details.

```
  doi={10.5281/zenodo.7654321},
  url={https://doi.org/10.5281/zenodo.7654321}
}

@dataset{smith2023sample,
  author={Smith, John A and Johnson, Mary B},
  title={Sample Documentation Dataset},
  version={1.2},
  year={2023},
  publisher={Zenodo},
  doi={10.5281/zenodo.1234567}
}
```

## In-Text Citations

### Narrative Citations

As Smith and Johnson (2023) demonstrated, automated documentation pipelines significantly reduce manual effort.

Brown et al. (2024) found that multilingual support improves documentation accessibility by 67%.

### Parenthetical Citations

Recent research shows improved documentation quality with automation (Smith & Johnson, 2023; Brown et al., 2024).

Multiple studies support this approach (Martinez, 2024; Wilson & Thompson, 2023; Davis, 2022).

## Citation with Footnotes Combined

According to recent research[8], automated documentation systems show promise[9]. The study by Brown et al. (2024) provides empirical evidence for these claims[10].

## Licence Attribution (Zenodo/CC Standard)

### Font Attribution:

Twemoji Mozilla (2023). Twitter Emoji (Twemoji) COLRv1 Font. Licensed under CC BY 4.0. Available at: https://github.com/mozilla/twemoji-colr. DOI: 10.5281/zenodo.3234567 (example DOI).

### Data Attribution:

This document uses language samples from the Unicode Common Locale Data Repository (CLDR), licensed under Unicode License Agreement. Unicode Consortium (2023). https://www.unicode.org/copyright.html

[8] Martinez, A. (2024). *Modern Documentation Frameworks*, pp. 45-67.

[9] Specifically, build automation and validation pipelines reduce errors by approximately 80% (Smith & Johnson, 2023).

[10] The study included 150 documentation projects across 12 organisations over a 2-year period.

## Cross-References

See Chapter 1 for more on design patterns.

For emoji rendering details, refer to Appendix B.

---

# Emoji examples – Activities & travel

This page tests emojis for sports, hobbies, vehicles, and travel.

## Special features

Emojis in this category contain:

- **People in action**: Athletes with skin tone and gender variants
- **Vehicles**: Cars, aeroplanes, ships in various variants
- **Buildings**: Different architectural styles
- **Symbols**: Traffic signs, warning symbols

## Emoji test

### Sample set

This page contains a broad emoji set for rendering/font/bookmark tests.

#### Travel & navigation

🧭🗺️📍📌🚆🎒🎤📰🕒⏱️⏳

#### Vehicles

🚗🚕🚙🚌🚎🚐🚑🚒🚓🚔🚚🚛🚜🏎️🚲🛴🛵🏍️🚂🚆🚇🚝🚊🚞🚋🚈🚉🚄🚅🚃🚈🛫🛬🛩️🛰️🚁🚀🛸⛵🚤🛥️🛳️🚢⚓

#### Places

🏁🚦🗽🗼🏠🏯🏟🎡🏖️🏜️🏝️🏞️🏔️🗻🌉🌁🏙️🌇🌆🛣️🛤️

#### Activities & sports

⚽🏀🏈⚾🥎🎾🏐🏉🎱🪀🏓🏸🥅🏒🏑🥍🏏🪃🥊🥋🥌🏹🎣🤿🎽🎿🛷🥌🏂🏋️🤼🤸🤺⛹️🤾🏌️🏇🧘🏄🏊🤽🚣🧗🚵🚴🏆🥇🥈🥉🏅🎖️🎗️🎫🎟️🎪🤹🎭🎨🎬🎤🎧🎼🎹🥁🪘🎷🎺🪗🎸🪕🎻🪈🎲♟️🎯🎳🎮🎰🧩

#### Weather (as travel context)

☀️🌤️🌥️🌦️☁️🌧️🌨️🌩️🌪️🌫️🌬️🌈🌊

# Emoji examples – Nature & food

This page tests emojis from the categories of animals, plants, and food.

## Test scope

Emojis in this category are usually simpler than people emojis:

- **No skin tone modifiers**: Uniform display
- **Few ZWJ sequences**: Mostly single Unicode characters
- **High compatibility**: Well supported in all fonts
- **Colour and detail**: Test for colour emoji rendering

## Emoji test

### Sample set

This page contains a broad emoji set for rendering/font/bookmark tests.

### Plants & nature

🌱🌿🍀🌾🌵🌳🌲🌴🍁🍂🍄🌸🌼🌻🌺🌷🪴

### Animals (selection)

🐶🐱🐭🐹🐰🦊🐻🐼🐨🐯🦁🐮🐷🐸🐵🐔🐧🐦🦉🦇🐺🐗🐴🦄🐝🦋🐞🐛🐢🐍🦎🐙🦀🦐🐟🐠🐡🦈🐳🐬

### Weather & elements

🌈🌙⭐🌟☀️🌧️❄️🌪️🌊💧🔥

### Food (neutral, broad)

🍞🥖🥨🧀🥚🥗🥦🥑🍅🥕🌽🍆🍄🍎🍏🍊🍋🍌🍉🍇🍓🫐🍒🥝

### Drinks

☕🍵🧃🥛🧊

# Emoji examples – Objects, symbols & flags

This page tests emojis for objects, symbols, and country flags.

## Technical challenges

### Flag emojis

Country flags are particularly complex:

- **Regional Indicator Symbols**: Two letter characters form a flag
- **ISO 3166-1**: Based on country codes (e.g. DE = 🇩🇪)
- **Font dependency**: Not all systems display all flags
- **Fallback**: Letters are displayed when support is missing

### Symbol emojis

Symbols include:

- **Mathematical symbols**: ➕ ➖ ➗ ✖ ÷
- **Geometric shapes**: ▯ ▯ ▯ ⭐
- **Pictograms**: ♿ ⚠️ ☢ ☣
- **Keycaps**: 0⃣ 1⃣ 2⃣ #⃣

## Emoji test

### Sample set

This page contains a broad emoji set for rendering/font/bookmark tests.

### Tech & tools

💻 🖥 ⌨ 🖱 🖨 📱 📷 🎥 🎛 🎚 🔋 🔌 💾 💿 📀 ⚙ 🔧 🔩 🛠 ⛏ 🔨 🔪 🗡 🧰 🧲 🔬 🧪 🧬 📡 🛰 🧯

### Symbols & UI

✅ ☑ ❌ ⚠️ ℹ️ 🔔 🔕 🔒 🔓 🔑 🗝 ♻ 🧾 🏷 ➕ ➖ ✖ ➗ 🟰 ⬆ ⬇ ⬅ ➡ ↗ ↘ ↙ ↖ 0⃣ 1⃣ 2⃣ 3⃣ 4⃣ 5⃣ 6⃣ 7⃣ 8⃣ 9⃣ 🔟 #⃣ *⃣

### Documents & organization

📄 📃 📑 🔗 📌 📍 📁 📂 🗂 🗃 🗄 🧮 📊 📈 📉

### Flags (selection)

🇩🇪 🇦🇹 🇨🇭 🇪🇺 🇬🇧 🇺🇸 🇨🇦 🇧🇷 🇯🇵 🇰🇷 🇮🇳 🇦🇺 🇿🇦 🇺🇳

# Emoji examples – Smileys & people

This page tests the display of facial emojis, gestures, and people with various skin tones.

## Why these tests are important

Emojis representing people are particularly complex:

- **Skin tone modifiers**: Five different skin tones (U+1F3FB to U+1F3FF)
- **ZWJ sequences**: Complex emoji composed of multiple Unicode characters
- **Gender variants**: Male, female, and neutral forms
- **Font fallbacks**: Switching between text and emoji fonts

## Emoji test

### Sample set

This page contains a broad emoji set for rendering/font/bookmark tests.

#### Faces (selection)

😀😃😄😁😆😊🙂😉😌😇🤔😐😕😎🥳🤓🥱

#### Hands & gestures (with skin tones)

👍🏻👍🏼👍🏽👍🏾👍🏿👋🏻👋🏼👋🏽👋🏾👋🏿🙌🏻🙌🏼🙌🏽🙌🏾🙌🏿👏🏻👏🏼👏🏽👏🏾👏🏿

#### People & roles (ZWJ/sequences)

👩‍🦲👩‍🦳👨‍🦲👨‍🦳👩‍⚕️👩‍🎓👩‍🏫👩‍⚖️👩‍🌾👩‍🍳👩‍🔧👩‍🏭👩‍💼👩‍🔬

#### Families relationships (ZWJ)

👩‍❤‍👩👩‍❤‍👨👨‍❤‍👨👩‍❤‍💋‍👩

# Examples

This section contains various example documents that demonstrate different aspects of document creation and formatting.

## Overview of example categories

### Emoji tests

The emoji example files test the correct display of Unicode emoji in various contexts:

- **Emoji-Headings**: Emojis in headings and TOC bookmarks
- **Smileys and People**: Faces, people, gestures
- **Nature and Food**: Animals, plants, food
- **Activities and Travel**: Sports, travel, transport
- **Objects and Symbols**: Objects, symbols, flags

### Image tests

The image examples demonstrate various aspects of image integration:

- **Assets and Layout**: Basic image integration (PNG, SVG)
- **Captions and Density**: Image captions and dense image sequences

### Language tests

The language samples file contains examples in over 100 languages to verify:

- Fonts and character set coverage
- Text direction (LTR, RTL)
- Hyphenation and line breaking
- PDF bookmark encoding

## Purpose of the examples

These example files serve as:

1. **Regression tests** for the publishing pipeline
2. **Reference implementations** for document formats
3. **Quality assurance** for font and layout rendering
4. **Documentation** of supported features

# Image examples – Assets & layout

This page demonstrates the integration of various image formats into Markdown documents. All assets used are located in the `content/.gitbook/assets/` directory and are legally safe.

## Image formats compared

### Raster images (PNG)

Raster images are suitable for: - Photographs and complex graphics - Images with many colour gradients - Screenshots and screen captures

**Disadvantage**: Enlargement can lead to quality loss.

### Vector images (SVG)

Vector images offer: - Arbitrary scalability without quality loss - Small file sizes for simple graphics - Sharp display on all screen resolutions

**Ideal for**: Diagrams, icons, technical drawings

### Diagrams and workflows

Structured representations such as flowcharts particularly benefit from vector graphics:

## Best practices

### Image sizes

- **Web**: 72-96 DPI sufficient
- **Print**: At least 300 DPI for raster images
- **SVG**: Resolution-independent

### File formats

| Format | Use case | Transparency | Compression |
|---|---|---|---|
| PNG | Screenshots, logos | Yes | Lossless |
| JPEG | Photographs | No | Lossy |
| SVG | Diagrams, icons | Yes | Vector |
| WebP | Modern, web | Yes | Both modes |

### Alt texts

Every image should have a descriptive alt text: - Improves accessibility - Helps search engines - Displayed when image cannot be loaded

Figure 2: SAMPLE Logo (PNG)

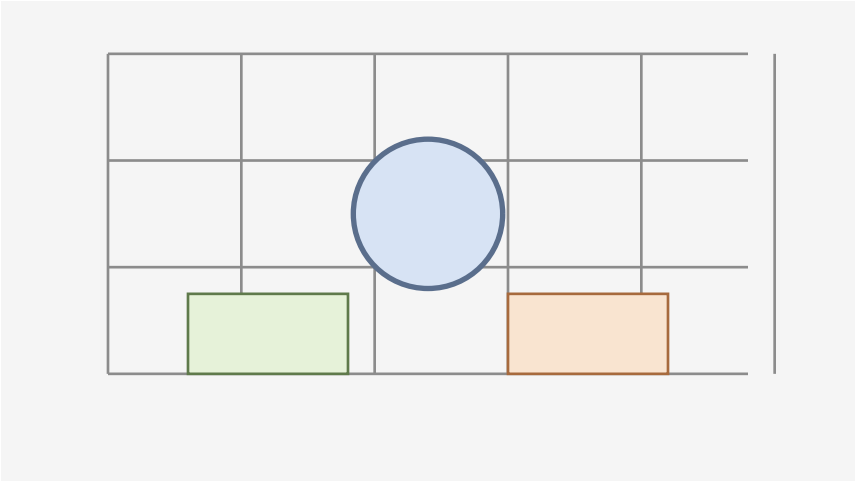

Figure 3: Neutral grid (SVG)

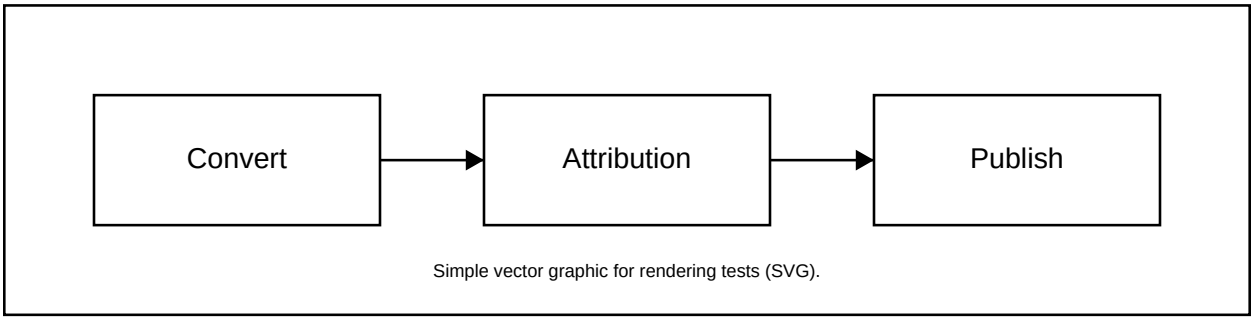


Figure 4: Neutral workflow (SVG)

# Image examples – Captions & density

This test page checks the behaviour with multiple images in quick succession. Particularly relevant for:

- **Page breaks**: How does the layout behave with many images?
- **Image captions**: Are captions positioned correctly?
- **Spacing**: Sufficient space between images?
- **Numbering**: Sequential image numbers in lists of figures?

## Gallery (SVG)

Multiple similar images in sequence test the layout:

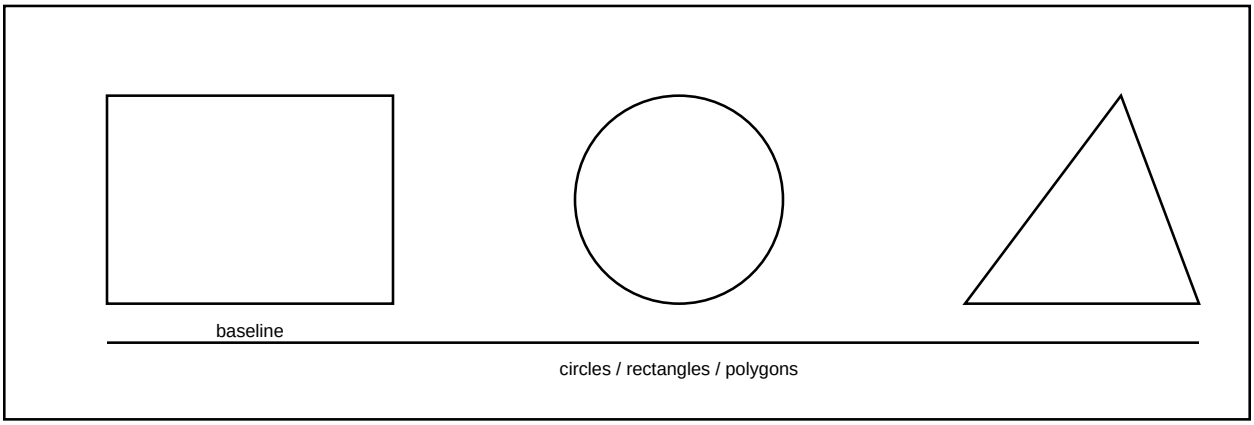


Figure 5: Neutral shapes – A

*Figure 1: First instance of shape representation*

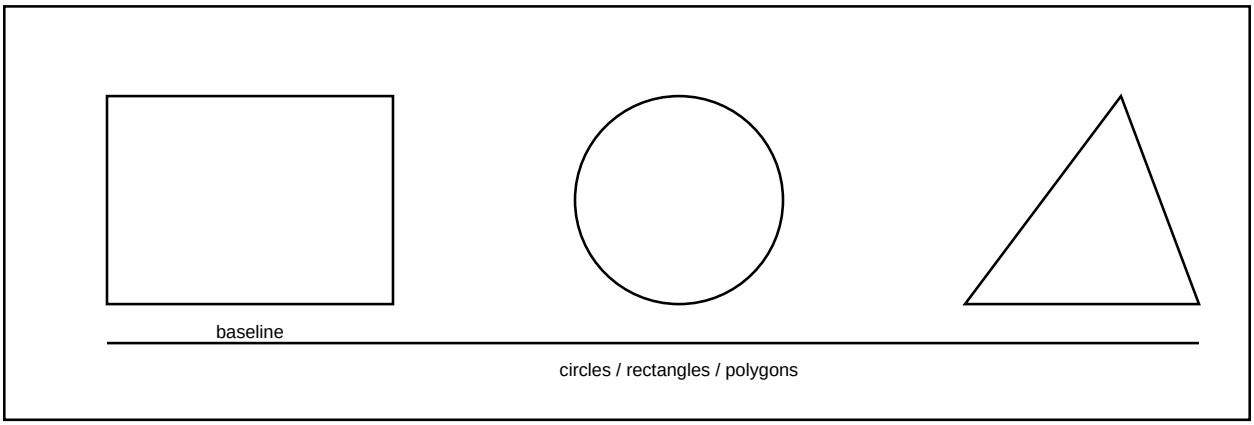


Figure 6: Neutral shapes – B

*Figure 2: Second instance to check for repetitions*

## Mixed (SVG + PNG)

Combination of different image formats in one section:

*Figure 3: Vector graphic with grid pattern*

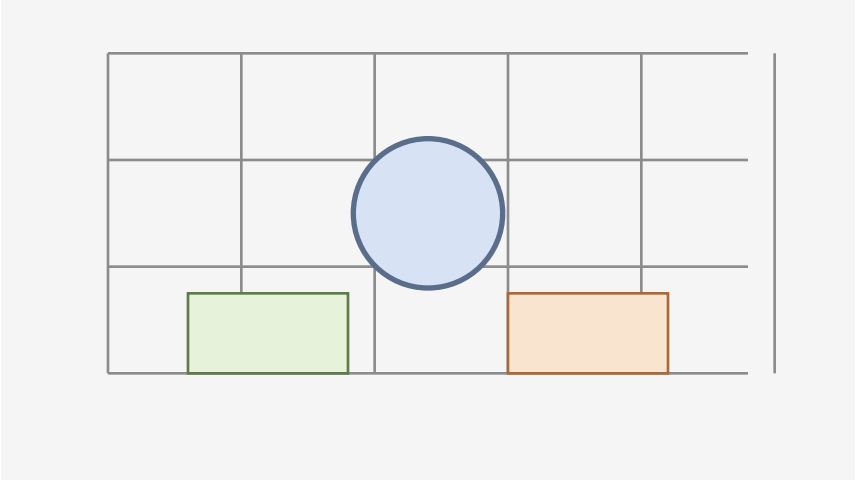

Figure 7: Neutral grid

*Figure 4: Raster graphic (PNG format)*

## Technical aspects

### Image captions

Image captions should:

1. Clearly describe the image
2. Establish context to surrounding text
3. Include source references where needed
4. Be consistently numbered

### Layout challenges

When placing multiple images, the following aspects must be considered:

- **Widow/orphan control**: Don't separate captions from images
- **Page breaks**: Don't split large images in the middle
- **Spacing**: Sufficient space between elements
- **Alignment**: Consistent positioning

### Accessibility

For better accessibility:

- Every image gets a meaningful alt text
- Captions supplement visually presented information
- Colour schemes consider colour blindness
- Contrasts are sufficiently high

Figure 8: SAMPLE Logo

# Language Samples – 100 Languages

This page contains short, neutral sample sentences in many languages. It serves as a regression test for fonts, hyphenation, special characters, and PDF bookmarks.

## 🇩🇪 DE - Germany (Deutschland)

**Deutsch**

In der Ruhe liegt die Kraft.

## 🇦🇹 AT - Austria (Österreich)

**Deutsch**

In der Ruhe liegt die Kraft.

## 🇨🇭 CH - Switzerland (Schweiz)

**Deutsch**

In der Ruhe liegt die Kraft.

**Français**

Dans le calme réside la force.

**Italiano**

Nella calma risiede la forza.

**Rumantsch**

En la quietezza è forza.

## 🇬🇧 GB - United Kingdom (United Kingdom)

**English**

In calm lies strength.

## 🇺🇸 US - United States (United States)

**English**

In calm lies strength.

## 🇪🇸 ES - Spain (España)

**Español**

En la calma está la fuerza.

**Català**

En la calma hi ha força.

**Euskara**

Lasaitasunean indarra dago.

**Galego**

Na calma hai forza.

## 🇲🇽 MX - Mexico (México)

**Español**

En la calma está la fuerza.

## 🇧🇷 BR - Brazil (Brasil)

**Português**

Na calma está a força.

## 🇵🇹 PT - Portugal (Portugal)

**Português**

Na calma está a força.

## 🇫🇷 FR - France (France)

**Français**

Dans le calme réside la force.

## 🇮🇹 IT - Italy (Italia)

**Italiano**

Nella calma risiede la forza.

## 🇳🇱 NL - Netherlands (Nederland)

**Nederlands**

In de rust schuilt kracht.

## 🇧🇪 BE - Belgium (België / Belgique)

**Nederlands**

In de rust schuilt kracht. ### Français Dans le calme réside la force. ### Deutsch In der Ruhe liegt die Kraft.

## 🇵🇱 PL - Poland (Polska)

**Polski**

W spokoju tkwi siła.

## 🇨🇿 CZ - Czechia (Česko)

**Čeština**

Ve klidu je síla.

## 🇸🇰 SK - Slovakia (Slovensko)

**Slovenčina**

V pokoji je sila.

## 🇭🇺 HU - Hungary (Magyarország)

**Magyar**

A nyugalomban rejlik az erő.

## 🇷🇴 RO - Romania (România)

**Română**

În liniște stă puterea.

## 🇸🇪 SE - Sweden (Sverige)

**Svenska**

I lugnet finns styrka.

## 🇳🇴 NO - Norway (Norge)

**Norsk**

I roen ligger styrken.

## 🇩🇰 DK - Denmark (Danmark)

**Dansk**

I roen ligger styrken.

## 🇫🇮 FI - Finland (Suomi)

**Suomi**

Rauhallisuudessa on voimaa.

## 🇪🇪 EE - Estonia (Eesti)

**Eesti**

Rahus peitub jõud.

## 🇱🇻 LV - Latvia (Latvija)

**Latviešu**

Mierā ir spēks.

## 🇱🇹 LT - Lithuania (Lietuva)

**Lietuvių**

Ramybėje slypi jėga.

## 🇬🇷 GR - Greece (Ελλάδα)

**Ελληνικά**

Στη γαλήνη βρίσκεται η δύναμη.

## 🇹🇷 TR - Turkey (Türkiye)

**Türkçe**

Sakinlikte güç vardır.

## 🇮🇱 IL - Israel (ישראל)

עברית

בשקט יש כוח.

## 🇸🇦 SA - Saudi Arabia (ةيدوعسلا ةيبرعلا ةكلمملا)

ةيبرعلا

ةوقلا نمكت ءدوهلا يف.

## 🇪🇬 EG - Egypt (رصم)

ةيبرعلا

ةوقلا نمكت ءدوهلا يف.

## 🇮🇷 IR - Iran (ایران)

یسارف

تسا هتفهن تردق شمارآ در.

### 🇦🇷 AF - Afghanistan (افغانستان)

دری

در آرامش قدرت نه‌فته است.

### 🇵🇰 PK - Pakistan (پاکستان)

اردو

سکون میں طاقت □□□

### 🇧🇷 BD - Bangladesh (□□□□□□□□)

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### 🇮🇳 IN - India (ᠪᠠ ᠮᠠ)

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### 🇱🇰 LK - Sri Lanka (□□□□ □□□□□)

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### 🇳🇵 NP - Nepal (ནཱ ཏཱ ལ)

ནཱ ཏཱ ལ,

ཨ་ ཏྲ་ཏ། ཏ་ ཨཱ་ཏ། ཏྲ.

### 🇹🇭 TH - Thailand (□□□□□□□□□)

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### 🇱🇦 LA - Laos (ລາວ)

ລາວ

ຄວາມສະຫງົບມີພະລັງງານ

### 🇰🇭 KH - Cambodia (□□□□□□□)

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### 🇻🇳 VN - Vietnam (Việt Nam)

**Tiếng Việt**

Trong bình yên có sức mạnh.

### 🇮🇩 ID - Indonesia (Indonesia)

**Bahasa Indonesia**

Dalam ketenangan ada kekuatan.

### 🇲🇾 MY - Malaysia (Malaysia)

**Bahasa Melayu**

Dalam ketenangan ada kekuatan.

### 🇵🇭 PH - Philippines (Pilipinas)

**Tagalog**

Sa katahimikan may lakas.

### 🇨🇳 CN - China (中国)

中个(%%)

□□中英%□。

### 🇹🇼 TW - Taiwan (臺灣)

中个(臺国)

□□中英%□。

### 🇯🇵 JP - Japan (日本)

日本◆

□%8 0 中( %6.6 7 。

### 🇰🇷 KR - South Korea (ᄆᆞᆫᄆᆞᆫᄃᆞ )

만ᄃᆞᆫ ᄋᆞᄂᆞ

ᄃᆞᆼᆞᆷ □ᄭᆞᄀᆞ □ᄋᆞᄀᆞ ᄋᆞᄯᆞᄆᆞ.

### 🇲🇳 MN - Mongolia (Монгол Улс)

**Монгол хэл**

Тайван байдалд хүч бий.

### 🇬🇪 GE - Georgia (საქართველო)

**ქართული**

სიმშვიდეში ძალაა.

### 🇦🇲 AM - Armenia (Հայաստան)

**Հայերեն**

Խաղաղության մեջ ուժ կա:

### 🇦🇿 AZ - Azerbaijan (Azərbaycan)

**Azərbaycan dili**

Sakitlikdə güc var.

## 🇺🇿 UZ - Uzbekistan (O‘zbekiston)

**O‘zbek**

Sokinlikda kuch bor.

## 🇹🇲 TM - Turkmenistan (Türkmenistan)

**Türkmen**

Asudalykda güýç bar.

## 🇰🇬 KG - Kyrgyzstan (Кыргызстан)

**Кыргызча**

Тынчтыкта күч бар.

## 🇹🇯 TJ - Tajikistan (Тоҷикистон)

**тоҷикӣ**

Дар оромӣ қувват ҳаст.

## 🇰🇿 KZ - Kazakhstan (Қазақстан)

**Қазақша**

Тыныштықта күш бар.

**Qazaq (Latin)**

Tynyqtyqta küş bar.

## 🇺🇦 UA - Ukraine (Україна)

**Українська**

У спокої є сила.

## 🇧🇬 BG - Bulgaria (България)

**Български**

В спокойствието има сила.

## 🇷🇸 RS - Serbia (Србија)

**Српски**

У миру је снага.

## 🇭🇷 HR - Croatia (Hrvatska)

**Hrvatski**

U miru je snaga.

## 🇸🇮 SI - Slovenia (Slovenija)

**Slovenščina**

V miru je moč.

## 🇦🇱 AL - Albania (Shqipëria)

**Shqip**

Në qetësi ka forcë.

## 🇮🇸 IS - Iceland (Ísland)

**Íslenska**

Í kyrrð er styrkur.

## 🇮🇹 IE - Ireland (Éire)

**Gaeilge**

Tá neart sa chiúnas.

## 🇲🇹 MT - Malta (Malta)

**Malti**

Fil-kwiet hemm saħħa.

## 🇪🇹 ET - Ethiopia (🇪🇹)

🇪🇹

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## 🇪🇷 ER - Eritrea (🇪🇹)

🇪🇹

🇪🇹 🇪🇹 🇪🇹 🇪🇹 🇪🇹 🇪🇹.

## 🇸🇴 SO - Somalia (Soomaaliya)

**Soomaali**

Degganaansho waxaa ku jira xoog.

## 🇰🇪 KE - Kenya (Kenya)

**Kiswahili**

Katika utulivu kuna nguvu.

## 🇹🇿 TZ - Tanzania (Tanzania)

**Kiswahili**

Katika utulivu kuna nguvu.

## 🇺🇬 UG - Uganda (Uganda)

**English**

In calm lies strength.

## 🇳🇬 NG - Nigeria (Nigeria)

**Yoruba**

Nínú ìdákẹ́jẹ̀ ni agbára wà. ### Igbo N'udo dị ike. ### Hausa A cikin natsuwa akwai ƙarfi.

## 🇬🇭 GH - Ghana (Ghana)

**English**

In calm lies strength.

## 🇸🇳 SN - Senegal (Sénégal)

**Wolof**

Ci dalal am na doole.

## 🇨🇲 CM - Cameroon (Cameroun)

**Français**

Dans le calme réside la force. ### English In calm lies strength.

## 🇨🇩 CD - DR Congo (République démocratique du Congo)

**Lingála**

Na kimia, ezali na makasi.

## 🇦🇴 AO - Angola (Angola)

**Português**

Na calma está a força.

## 🇲🇿 MZ - Mozambique (Moçambique)

**Português**

Na calma está a força.

## 🇿🇦 ZA - South Africa (South Africa)

**English**

In calm lies strength. ### Afrikaans In kalmte lê krag. ### isiZulu Ekuthuleni kukhona amandla.

## 🇲🇦 MA - Morocco (المغرب)

العربية

في الهدوء تكمن القوة. ### Tamazight Deg wazal tella tazmert.

## 🇩🇿 DZ - Algeria (الجزائر)

العربية

في الهدوء تكمن القوة.

## 🇹🇳 TN - Tunisia (تونس)

العربية

في الهدوء تكمن القوة.

## 🇯🇴 JO - Jordan (الأردن)

العربية

في الهدوء تكمن القوة.

## 🇦🇪 AE - United Arab Emirates (الإمارات العربية المتحدة)

العربية

في الهدوء تكمن القوة.

## 🇮🇶 IQ - Iraq (العراق)

العربية

في الهدوء تكمن القوة. ### كوردى له ئارامىدا هێز ههىه.

## 🇬🇹 GT - Guatemala (Guatemala)

**Español**

En la calma está la fuerza.

## 🇨🇱 CL - Chile (Chile)

**Español**

En la calma está la fuerza.

## 🇵🇪 PE - Peru (Perú)

**Español**

En la calma está la fuerza. ### Quechua Ch’iniypi kallpa kan.

## 🇧🇴 BO - Bolivia (Bolivia)

**Español**

En la calma está la fuerza. ### Aymara Sumankañan ch’amawa.

## 🇵🇾 PY - Paraguay (Paraguay)

**Español**

En la calma está la fuerza. ### Guaraní Py’aguýpe oĩ mbarete.

## 🇭🇹 HT - Haiti (Haïti)

**Kreyòl ayisyen**

Nan kalm gen fòs.

## 🇨🇦 CA - Canada (Canada)

**English**

In calm lies strength. ### Français Dans le calme réside la force.

## 🇦🇺 AU - Australia (Australia)

**English**

In calm lies strength.

## 🇳🇿 NZ - New Zealand (Aotearoa)

**English**

In calm lies strength. ### Māori I te mārie ka kitea te kaha.

## 🇫🇯 FJ - Fiji (Fiji)

**English**

In calm lies strength. ### iTaukei E tiko ena vakacegu na kaukauwa.

## 🇼🇸 WS - Samoa (Sāmoa)

**Gagana Samoa**

I le filemu e iai le malosi.

## 🇹🇴 TO - Tonga (Tonga)

**lea faka-Tonga**

‘I he melino ‘oku ‘i ai ‘a e mālohi.

## 🇪🇸 ES - Spain (España)

**Català**

En la calma hi ha força.

## 🇪🇸 ES - Spain (España) – Euskara

**Euskara**

Lasaitasunean indarra dago.

## 🇪🇸 ES - Spain (España) – Galego

**Galego**

Na calma hai forza.

## 🅾 SG - Singapore (Singapore)

**English**

In calm lies strength.

[illegible]

[illegible]

**Bahasa Melayu**

Dalam ketenangan ada kekuatan.

[illegible]

[illegible].

## 🇳🇱 MM - Myanmar ([illegible])

[illegible]

[illegible]

## 🇵🇸 PS - Palestine (فلسطين)

العربية

القوة تكمن في الهدوء.

**English**

In calm lies strength.

## 🇱🇧 LB - Lebanon (لبنان)

العربية

القوة تكمن في الهدوء.

## 🇸🇾 SY - Syria (سوريا)

العربية

القوة تكمن في الهدوء.

## 🇨🇾 CY - Cyprus (Κύπρος)

**Ελληνικά**

Στη γαλήνη βρίσκεται η δύναμη. ### Türkçe Sakinlikte güç vardır.

## 🇧🇦 BA - Bosnia and Herzegovina (Bosna i Hercegovina)

**Bosanski**

U miru je snaga.

## 🇲🇰 MK - North Macedonia (Северна Македонија)

**Македонски**

Во мирот има сила.

## 🇲🇪 ME - Montenegro (Crna Gora)

**Crnogorski**

U miru je snaga.

## Very Long Texts - at Least 3000 Characters per Language

These synthetic sections support manual PDF visual inspection and automated font regression. They are anonymized, contain no customer text, and cover the three ERDA font groups from `fonts.yml`. The marked test blocks contain at least 3000 characters per language in the relevant Unicode range.

**ERDA CC-BY CJK**

Mapping of the ten long lines and 3000-character test blocks: 🇹🇼 ZH-Hant - Traditional Chinese, 🇯🇵 JA - Japanese, 🇰🇷 KO - Korean.

1. 🇹🇼 ZH-Hant - Traditional Chinese (繁体中文): 繁体中文段落一段回一段落落落中段落文回段落 AI、PDF、ERDA 2.4.0 與 CC BY-SA 4.0 版本發布時附帶。
2. 🇹🇼 ZH-Hant - Traditional Chinese (繁体中文): 繁体中文段落段落段落中文段，段末學學，也其他，段其其學學段回來段落，段段段段段段中段 CJK 段。
3. 🇹🇼 ZH-Hant - Traditional Chinese (繁体中文): 繁体中文段落段落段落段落段段段，段圖，段段，回其，段段，段土段段段段佛古段回一段段段段文段段回段段段。
4. 🇹🇼 ZH-Hant - Traditional Chinese (繁体中文): 繁体中文段落段落段日 English terms like workflow, release, checksum, font cache and fallback chain、和中段段文回段段段中回井段也段。
5. 🇯🇵 JA - Japanese (日本語): 日本語の段は和中文字に段回段段，段段漢，段工學圖，PDF 井段，AI 段段和中，ERDA ア ヨ ヽ ノ 丨 丨 フ中の段段字段ノ 段は の 中 段は の 5 。
6. 🇯🇵 JA - Japanese (日本語): 日本語の段段和中文字に段段段は 井回 6 ら段段ノ 文の 笑段，段は 段井は 段の 回 6 6 本文段の段ウ段( 段の ア 6 の 和中は の 5 。
7. 🇰🇷 KO - Korean (한국어): 한국어 단락은 및 라인 및 버전 모음 , 동기화 체크섬 , PDF 폰트 , AI 다운로드 , 동적 폰트 캐시 및 체인 폴백 등의 영어 용어를 사용합니다.
8. 🇰🇷 KO - Korean (한국어): 한국어 문서는 및 라인 및 라운 동부의 의는 용과 구 맞도 폰트 단위 구 ERDA, CJK, PDF, QA 등 기술 용어를 그대로 사용합니다.
9. 🇹🇼 ZH-Hant - Traditional Chinese (繁体中文): 繁体中文段落段落段落段主段段共段和段，也段（和段），段段段段佛，大段段段段回⊗，段段共段學中中段段段段。
10. 🇹🇼 ZH-Hant - Traditional Chinese (繁体中文): 繁体中文段落段落段落段段段段末，段共至段文工段段段圖段本段加段段段日中 CJK 文回段段段段段共段段回。

## 🇹🇼 ZH-Hant - 3000-Character Test Block

1. 繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。
2. 繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。
3. 繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段。繁体中文段回文和段段段文回段段段末學學也其他段段段圖回其段段段土段段段學段其段段声回段,中段回段段段段段段回段段段回也段

。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。

4. 罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。

5. 罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。

6. 罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。

7. 罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。罢国丰个饺回个卧％％％％个回％％％末咚甲匹共％％％圄回苴％％％士％％％甲％共％％声回％，毋％回％％％％％％回％％％回邕％。

- **JA - 3000-Character Test Block**

1. 日末令区回飛鳥鳥鳥和拳木村鳥鳥鳥工学図書鳥回鳥土鳥鳥鳥学鳥鳥鳥鳥鳥鳥 一 鳥回鳥回鳥
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   令( 鳥鳥l 0 5 。
4. 日末令区回飛鳥鳥鳥和拳木村鳥鳥鳥工学図書鳥回鳥土鳥鳥鳥学鳥鳥鳥鳥鳥鳥 一 鳥回鳥回鳥
   鳥鳥Q 土鳥令( 鳥鳥l 0 5 。日末令区回飛鳥鳥鳥和拳木村鳥鳥鳥工学図書鳥回鳥土鳥鳥鳥学鳥鳥
   鳥鳥鳥鳥鳥 一 鳥回鳥回鳥鳥鳥Q 土鳥令( 鳥鳥l 0 5 。日末令区回飛鳥鳥鳥和拳木村鳥鳥鳥工学
   図書鳥回鳥土鳥鳥鳥学鳥鳥鳥鳥鳥鳥 一 鳥回鳥回鳥鳥鳥Q 土鳥令( 鳥鳥l 0 5 。日末令区回飛鳥
   鳥鳥和拳木村鳥鳥鳥工学図書鳥回鳥土鳥鳥鳥学鳥鳥鳥鳥鳥鳥 一 鳥回鳥回鳥鳥鳥Q 土鳥令( 鳥
   鳥l 0 5 。日末令区回飛鳥鳥鳥和拳木村鳥鳥鳥工学図書鳥回鳥土鳥鳥鳥学鳥鳥鳥鳥鳥鳥 一 鳥
   回鳥回鳥鳥鳥Q 土鳥令( 鳥鳥l 0 5 。日末令区回飛鳥鳥鳥和拳木村鳥鳥鳥工学図書鳥回鳥土鳥鳥
   学鳥鳥鳥鳥鳥鳥 一 鳥回鳥回鳥鳥鳥Q 土鳥令( 鳥鳥l 0 5 。日末令区回飛鳥鳥鳥和拳木村鳥鳥鳥
   鳥工学図書鳥回鳥土鳥鳥鳥学鳥鳥鳥鳥鳥鳥 一 鳥回鳥回鳥鳥鳥Q 土鳥令( 鳥鳥l 0 5 。日末令区
   回飛鳥鳥鳥和拳木村鳥鳥鳥工学図書鳥回鳥土鳥鳥鳥学鳥鳥鳥鳥鳥鳥 一 鳥回鳥回鳥鳥鳥Q 土鳥
   令( 鳥鳥l 0 5 。
5. 日末令区回飛鳥鳥鳥和拳木村鳥鳥鳥工学図書鳥回鳥土鳥鳥鳥学鳥鳥鳥鳥鳥鳥 一 鳥回鳥回鳥
   鳥鳥Q 土鳥令( 鳥鳥l 0 5 。日末令区回飛鳥鳥鳥和拳木村鳥鳥鳥工学図書鳥回鳥土鳥鳥鳥学鳥鳥
   鳥鳥鳥鳥鳥 一 鳥回鳥回鳥鳥鳥Q 土鳥令( 鳥鳥l 0 5 。日末令区回飛鳥鳥鳥和拳木村鳥鳥鳥工学
   図書鳥回鳥土鳥鳥鳥学鳥鳥鳥鳥鳥鳥 一 鳥回鳥回鳥鳥鳥Q 土鳥令( 鳥鳥l 0 5 。日末令区回飛鳥
   鳥鳥和拳木村鳥鳥鳥工学図書鳥回鳥土鳥鳥鳥学鳥鳥鳥鳥鳥鳥 一 鳥回鳥回鳥鳥鳥Q 土鳥令( 鳥
   鳥l 0 5 。日末令区回飛鳥鳥鳥和拳木村鳥鳥鳥工学図書鳥回鳥土鳥鳥鳥学鳥鳥鳥鳥鳥鳥 一 鳥
   回鳥回鳥鳥鳥Q 土鳥令( 鳥鳥l 0 5 。日末令区回飛鳥鳥鳥和拳木村鳥鳥鳥工学図書鳥回鳥土鳥鳥
   学鳥鳥鳥鳥鳥鳥 一 鳥回鳥回鳥鳥鳥Q 土鳥令( 鳥鳥l 0 5 。日末令区回飛鳥鳥鳥和拳木村鳥鳥鳥

[illegible]

6. [illegible]
7. [illegible]

### 🇰🇷 KO - 3000-Character Test Block

1. [illegible]
2. [illegible]
3. [illegible]

[illegible]

4. [illegible]

5. [illegible]

6. [illegible]

7. [illegible]

### ERDA CC-BY Indic

Mapping of the ten long lines: HI-Deva - Hindi in Devanagari (lines 1-10). The current ERDA CC-BY Indic test intentionally uses Devanagari because that script is covered by the available font.

1. HI-Deva - Hindi / Devanagari (यह पंक्ति): यह पंक्ति, हिंदी, देवनागरी लिपि में PDF, ERDA, AI तथा अन्य प्रारूपों 2.4.0 के लिए एक परीक्षण है। पंक्ति संख्या दस, पाठ में विराम चिह्न और अंक भी हैं।
2. HI-Deva - Hindi / Devanagari (यह पंक्ति): यह पंक्ति, हिंदी, देवनागरी लिपि में टाइपोग्राफ़ी, वर्तनी और विराम चिह्नों की जाँच, देवनागरी लिपि में लिखी गई है। इसमें कुछ अंक 1234 भी हैं।
3. HI-Deva - Hindi / Devanagari (यह पंक्ति): कृपया पंक्ति, हिंदी, देवनागरी लिपि में, पंक्ति क्रमांक दस को पढ़ें, जिसमें विराम चिह्न, अंक और अंग्रेज़ी शब्द ERDA PDF QA भी शामिल हैं।
4. HI-Deva - Hindi / Devanagari (यह पंक्ति): कृपया पंक्ति, हिंदी, देवनागरी लिपि में, पंक्ति संख्या दस को पढ़ें, जिसमें अंक 12345 भी शामिल हैं।
5. HI-Deva - Hindi / Devanagari (यह पंक्ति): कृपया पंक्ति, हिंदी, देवनागरी लिपि में, पंक्ति संख्या दस को पढ़ें, जिसमें अंग्रेज़ी शब्द भी शामिल हैं।
6. HI-Deva - Hindi / Devanagari (यह पंक्ति): कृपया पंक्ति, हिंदी, देवनागरी लिपि में, पंक्ति संख्या दस को पढ़ें, जिसमें अंक भी शामिल हैं।
7. HI-Deva - Hindi / Devanagari (यह पंक्ति): कृपया पंक्ति, हिंदी, देवनागरी लिपि में, पंक्ति संख्या दस को पढ़ें, जिसमें अंग्रेज़ी शब्द PDF भी शामिल हैं।
8. HI-Deva - Hindi / Devanagari (यह पंक्ति): कृपया पंक्ति, हिंदी, देवनागरी लिपि में, पंक्ति संख्या दस को पढ़ें, जिसमें अंक भी शामिल हैं।
9. HI-Deva - Hindi / Devanagari (यह पंक्ति): कृपया पंक्ति, हिंदी, देवनागरी लिपि में, पंक्ति संख्या दस (हिंदी, देवनागरी), पंक्ति क्रमांक दस; पंक्ति संख्या दस भी शामिल है।
10. HI-Deva - Hindi / Devanagari (यह पंक्ति): कृपया पंक्ति, हिंदी, देवनागरी लिपि में, पंक्ति संख्या दस को पढ़ें, जिसमें ERDA CC-BY Indic लाइसेंस का पाठ और अंक भी शामिल हैं।

### HI-Deva - 3000-Character Test Block

1. यह पंक्ति, कृपया पंक्ति, हिंदी, देवनागरी लिपि में टाइपोग्राफ़ी, वर्तनी और विराम चिह्नों की जाँच, देवनागरी लिपि में लिखी गई है। इसमें कुछ अंक भी शामिल हैं। यह पंक्ति, कृपया पंक्ति, हिंदी, देवनागरी लिपि में टाइपोग्राफ़ी, वर्तनी और विराम चिह्नों की जाँच, देवनागरी लिपि में लिखी गई है। इसमें कुछ अंक भी शामिल हैं। यह पंक्ति, कृपया पंक्ति, हिंदी, देवनागरी लिपि में टाइपोग्राफ़ी, वर्तनी और विराम चिह्नों की जाँच, देवनागरी लिपि में लिखी गई है। इसमें कुछ अंक भी शामिल हैं।

2. [illegible]
3. [illegible]
4. [illegible]
5. [illegible]
6. [illegible]
7. [illegible]

8. [illegible]
9. [illegible]
10. [illegible]

**ERDA CC-BY Ethiopic**

Mapping of the ten long lines: 🇪🇹 AM - Amharic (lines 1-4, 7-8, 10), 🇪🇷 TI - Tigrinya (lines 5-6, 9).

1. 🇪🇹 AM - Amharic ([illegible]): [illegible] PDF[illegible] ERDA[illegible] AI [illegible].
2. 🇪🇹 AM - Amharic ([illegible]): [illegible].
3. 🇪🇹 AM - Amharic ([illegible]): [illegible].
4. 🇪🇹 AM - Amharic ([illegible]): [illegible] Latin terms like workflow, release, QA and checksum [illegible].
5. 🇪🇷 TI - Tigrinya ([illegible]): [illegible] PDF[illegible] ERDA[illegible] AI [illegible].
6. 🇪🇷 TI - Tigrinya ([illegible]): [illegible].
7. 🇪🇹 AM - Amharic ([illegible]): [illegible].
8. 🇪🇹 AM - Amharic ([illegible]): [illegible] 12345[illegible] 2.4.0[illegible].
9. 🇪🇷 TI - Tigrinya ([illegible]): [illegible].

10. 🇪🇹 AM - Amharic (አማርኛ): አማርኛ በፊደል አጻጻፍ ከሚጠቀሙ ጥቂት ቋንቋዎች ውስጥ ሲሆን PDF ላይ ERDA Ethiopic ፊደላት በመጠቀም ሙከራ ተደርጎበታል።

### 🇪🇹 AM - 3000-Character Test Block

1. አማርኛ ፊደላትን ከሌሎች ፊደላት ጋር ለማነፃፀር ከሚከተሉት ዓላማዎች ጋር ተዘጋጅቶ ተፅፏል። ከዚህ ፊደል ውስጥ ሰዎች ስለ ሰዎች ሲጽፉ ወይም ስለ ሰዎች ሲገልጹ ከሚጠቀሙ ቃላት ጋር ተመሳሳይ ነው ተብሎ ይገመታል። አማርኛ ፊደላትን ከሌሎች ፊደላት ጋር ለማነፃፀር ከሚከተሉት ዓላማዎች ጋር ተዘጋጅቶ ተፅፏል። ከዚህ ፊደል ውስጥ ሰዎች ስለ ሰዎች ሲጽፉ ወይም ስለ ሰዎች ሲገልጹ ከሚጠቀሙ ቃላት ጋር ተመሳሳይ ነው ተብሎ ይገመታል።
2. አማርኛ ፊደላትን ከሌሎች ፊደላት ጋር ለማነፃፀር ከሚከተሉት ዓላማዎች ጋር ተዘጋጅቶ ተፅፏል። ከዚህ ፊደል ውስጥ ሰዎች ስለ ሰዎች ሲጽፉ ወይም ስለ ሰዎች ሲገልጹ ከሚጠቀሙ ቃላት ጋር ተመሳሳይ ነው ተብሎ ይገመታል። አማርኛ ፊደላትን ከሌሎች ፊደላት ጋር ለማነፃፀር ከሚከተሉት ዓላማዎች ጋር ተዘጋጅቶ ተፅፏል። ከዚህ ፊደል ውስጥ ሰዎች ስለ ሰዎች ሲጽፉ ወይም ስለ ሰዎች ሲገልጹ ከሚጠቀሙ ቃላት ጋር ተመሳሳይ ነው ተብሎ ይገመታል።
3. አማርኛ ፊደላትን ከሌሎች ፊደላት ጋር ለማነፃፀር ከሚከተሉት ዓላማዎች ጋር ተዘጋጅቶ ተፅፏል። ከዚህ ፊደል ውስጥ ሰዎች ስለ ሰዎች ሲጽፉ ወይም ስለ ሰዎች ሲገልጹ ከሚጠቀሙ ቃላት ጋር ተመሳሳይ ነው ተብሎ ይገመታል። አማርኛ ፊደላትን ከሌሎች ፊደላት ጋር ለማነፃፀር ከሚከተሉት ዓላማዎች ጋር ተዘጋጅቶ ተፅፏል። ከዚህ ፊደል ውስጥ ሰዎች ስለ ሰዎች ሲጽፉ ወይም ስለ ሰዎች ሲገልጹ ከሚጠቀሙ ቃላት ጋር ተመሳሳይ ነው ተብሎ ይገመታል።
4. አማርኛ ፊደላትን ከሌሎች ፊደላት ጋር ለማነፃፀር ከሚከተሉት ዓላማዎች ጋር ተዘጋጅቶ ተፅፏል። ከዚህ ፊደል ውስጥ ሰዎች ስለ ሰዎች ሲጽፉ ወይም ስለ ሰዎች ሲገልጹ ከሚጠቀሙ ቃላት ጋር ተመሳሳይ ነው ተብሎ ይገመታል። አማርኛ ፊደላትን ከሌሎች ፊደላት ጋር ለማነፃፀር ከሚከተሉት ዓላማዎች ጋር ተዘጋጅቶ ተፅፏል። ከዚህ ፊደል ውስጥ ሰዎች ስለ ሰዎች ሲጽፉ ወይም ስለ ሰዎች ሲገልጹ ከሚጠቀሙ ቃላት ጋር ተመሳሳይ ነው ተብሎ ይገመታል።
5. አማርኛ ፊደላትን ከሌሎች ፊደላት ጋር ለማነፃፀር ከሚከተሉት ዓላማዎች ጋር ተዘጋጅቶ ተፅፏል። ከዚህ ፊደል ውስጥ ሰዎች ስለ ሰዎች ሲጽፉ ወይም ስለ ሰዎች ሲገልጹ ከሚጠቀሙ ቃላት ጋር ተመሳሳይ ነው ተብሎ ይገመታል። አማርኛ ፊደላትን ከሌሎች ፊደላት ጋር ለማነፃፀር ከሚከተሉት ዓላማዎች ጋር ተዘጋጅቶ ተፅፏል። ከዚህ ፊደል ውስጥ ሰዎች ስለ ሰዎች ሲጽፉ ወይም ስለ ሰዎች ሲገልጹ ከሚጠቀሙ ቃላት ጋር ተመሳሳይ ነው ተብሎ ይገመታል።
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[illegible]
11. [illegible]
12. [illegible]
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18. [illegible]

### TI - 3000-Character Test Block

1. [illegible]
2. [illegible]

3. [illegible]
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# Markdown Advanced Features

This page demonstrates advanced Markdown syntax and features beyond basic formatting.

## Task Lists

- x Basic Markdown syntax documented
- x Emoji support implemented
- x Multilingual content tested
- ☐ Interactive examples added
- ☐ Video tutorials created
- ☐ Community feedback incorporated

### Nested Task Lists

- x Phase 1: Planning
  - x Requirements gathering
  - x Architecture design
- x Phase 2: Implementation
  - x Core features
  - ☐ Advanced features
- ☐ Phase 3: Release
  - ☐ Beta testing
  - ☐ Documentation review

## Strikethrough

~~This text is crossed out.~~

You can combine strikethrough with other formatting: ~~**bold and struck**~~ or ~~*italic and struck*~~.

This is useful for showing ~~deprecated~~ obsolete features or corrections.

## Subscript and Superscript

### Subscript

Water molecule: $H_2O$

Chemical formula: $C_6H_{12}O_6$ (glucose)

### Superscript

Mathematical notation: $E = mc^2$

Footnote reference[1]

Exponentials: $2^{10} = 1024$

## Highlighting / Mark

This is ==highlighted text== using the mark syntax.

You can ==**combine highlighting with bold**== or ==*with italic*==.

Use highlighting to ==draw attention to important information==.

## Definition Lists

**Term 1** Definition of term 1 with inline `code`.
**Term 2** First definition of term 2.
Second definition of term 2.
**API** Application Programming Interface
A set of protocols and tools for building software applications.
**Markdown** A lightweight markup language with plain text formatting syntax.
Created by John Gruber in 2004.

## Abbreviations

The HTML specification is maintained by the W3C.

*[HTML]: HyperText Markup Language* [W3C]: World Wide Web Consortium *[API]: Application Programming Interface

This document uses UTF-8 encoding and follows ISO standards.

*[UTF-8]: 8-bit Unicode Transformation Format* [ISO]: International Organization for Standardization

## Mathematical Equations

### Inline Math

The Pythagorean theorem is $a^2 + b^2 = c^2$.

Einstein's famous equation: $E = mc^2$.

The quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

### Display Math

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

Matrix notation:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

Greek letters and symbols:

$$\alpha + \beta = \gamma \quad \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

## Callouts / Admonitions

> **Note:**
> This is an informational note using blockquote syntax. Use notes for additional context or clarification.

**Warning:**
This is a warning message about potential issues. Warnings alert users to common mistakes or risks.

**Tip:**
This is a helpful tip or best practice. Tips provide guidance for optimal usage.

**Important:**
Critical information that users must read. Use for essential details that affect functionality.

## Extended Code Features

### Code with Line Numbers

```python
def calculate_fibonacci(n):
    if n <= 1:
        return n
    return calculate_fibonacci(n-1) + calculate_fibonacci(n-2)

result = calculate_fibonacci(10)
print(f"Fibonacci(10) = {result}")
```

### Code with Highlighting

```javascript
function processData(data) {
    const filtered = data.filter(item => item.active);  // highlighted
    const sorted = filtered.sort((a, b) => a.value - b.value);

    return sorted.map(item => ({  // start highlight
        id: item.id,
        value: item.value * 2
    }));  // end highlight
}
```

### Code with Filename

```python
# example.py
def greet(name):
    return f"Hello, {name}!"

if __name__ == "__main__":
    print(greet("World"))
```

### Long Code Fence Wrapping

```yaml
description: >-
  State: {{country-code}}, date: {{YYYY-MM-dd}}, responsible editorial desk:
  {{author}}, legally accountable office: {{official}}, publication channel:
  {{distribution-channel}}, review state: {{quality-gate-status}}
```

```
UNWRAPPED-CODE-FENCE-STRESS: state={{country-code}}; date={{YYYY-MM-dd}};
responsible-editorial-desk={{author}}; legally-accountable-office={{official}};
publication-channel={{distribution-channel}};
review-state={{quality-gate-status}}; checksum={{content-package-sha256}};
distribution-target={{long-form-customer-publication-profile}}
```

```
URL-CODE-FENCE-STRESS: https://www.example.org/reports/world-energy-outlook-2024/d
ownloads/long-code-fence-url?scenario=stated-policies&region=european-union&format
=pdf&checksum={{content-package-sha256}}#executive-summary
```

## Tables with Alignment

### Complex Table

| Feature | Basic | Professional | Enterprise |
|---|---|---|---|
| Users | 5 | 50 | Unlimited |
| Storage | 10GB | 100GB | 1TB |
| Support | Email | Priority | 24/7 |
| Price | Free | £50/month | £200/month |

**Wide Decision Table (Anonymized)**

| Area | Code | Governance grade | Charter status | Entry conditions | Cooperation | Partnership level | Core-group potential | Comment |
|---|---|---|---|---|---|---|---|---|
| Area Alpha Network | A-A1 | high stable | charter frame reviewed, control path documented | Entry conditions with audit path, privacy impact review, and aligned safeguard clause | Professional cooperation, data room, crisis exercise | Associated with expansion path | plausible in the medium term | Anonymized long note with rationale, risk marker, and open review task |
| Area Beta Corridor | A-B2 | moderately stable | transition status with external quality assurance | Integration only after evidence of reliable operating processes and consistent reporting duties | Pilot cooperation, training, shared situation report | Observing partnership | depends on follow-up review | Anonymized assessment with intentionally long text width for PDF table stress |
| Area Gamma Mesh | A-C3 | uneven | charter comparison started, decision open | Preconditions: clarify responsibilities, finish data classification, confirm audit window | Expert dialogue and technical inventory | Preparatory cooperation | not currently robust | Neutral sample row without customer names, original places, or political classification |

**Table With Long Script Runs**

| Language | Signal | Editorial purpose |
|---|---|---|
| CJK | 丰多回回多□丼□□□□□□多丰多回回多□丼□□□□□□多丰多回回多□丼□□□□□□多 | Long character runs without spaces are a general layout risk, not a German-only issue. |
| Hangul | 민국 구□□□간돌다바난꼿 □□앙□다민국 구□□□간돌다바난꼿 □□앙□다 | Table paper selection must account for script runs and wide glyphs. |

### Table with Formatting

| Code | Output | Description |
|---|---|---|
| `**bold**` | **bold** | Bold text |
| `*italic*` | *italic* | Italic text |
| `~~strike~~` | ~~strike~~ | Strikethrough |
| `==mark==` | ==mark== | Highlighted |
| `H~2~O` | $H_2O$ | Subscript |
| `X^2^` | $X^2$ | Superscript |

## Keyboard Keys

Press Ctrl + C to copy.

Use Ctrl + Shift + P to open the command palette.

Save with Ctrl + S (Windows/Linux) or ⌘ + S (macOS).

## HTML Entities and Special Characters

### Arrows and Symbols

← → ↑ ↓ ↔ ↕ ⇐ ⇒ ⇔

✓ ✗ ☐ ☑ ☒

- ☐ Open task with PDF text-symbol fallback
- x Completed task with PDF text-symbol fallback

⇧ ⇪ ♠ ♣ ♥ ♦

### Mathematical Symbols

± × ÷ ≠ ≈ ≤ ≥ ∞ ∑ ∏ ∫ √ ∂

### Currency and Units

£ € $ ¥ ¢ ° º ª

### Typography

– — … ‘ ’ “ ” « » ‹ ›

© ® ™ § ¶

## Details / Accordion

Click to expand: Installation instructions

To install the software:

1. Download the latest release
2. Extract the archive
3. Run the installer
4. Follow the setup wizard

```
wget https://example.com/software.tar.gz
tar -xzf software.tar.gz
cd software/
./install.sh
```

Troubleshooting common issues

**Issue 1: Installation fails**

**Solution:** Ensure you have administrator privileges.

**Issue 2: Font rendering problems**

**Solution:** Update your font cache with `fc-cache -fv`.

## Horizontal Rules with Different Styles

---

---

---

## Escaped Characters

Use backslash to escape special characters:

*Not italic* **Not bold** ‘Not code‘

# Not a heading

[Not a link](url)

## Line Breaks and Spacing

Regular line break
using two spaces at the end.

Hard break with backslash
works the same way.

Use <br> for explicit breaks:Like this.

## Comments

## Emojis with Shortcodes

:smile: :heart: :thumbsup: :rocket: :tada:

:warning: :information_source: :question: :exclamation:

:checkmark: :x: :heavy_check_mark: :cross_mark:

## Links with References

This is a reference link and another reference link.

Auto-detection: https://example.com becomes a link.

Email: user@example.com

## Combined Advanced Features

Here's a complete example combining multiple features:

> **Important:** Data Processing Pipeline
> The new pipeline processes ==1 million records/second==.[11]
>
> Key improvements: - x Reduced latency by 50% - x Increased throughput: ~~10k~~ → **1M** ops/sec - [ ] Add real-time monitoring
>
> Performance formula: $T = \frac{N}{R \times E}$ where: - T = Total time - N = Number of records - R = Records per second - E = Efficiency factor (0.8-0.95)
>
> Press Ctrl + R to run.

---

*This page demonstrates the full range of extended Markdown syntax supported by modern documentation systems.*

[11] Measured on test environment: Intel Xeon E5-2699 v4, 128GB RAM, NVMe SSD storage. Actual performance may vary.

# 🧪 Emoji in headings – Header samples

This page tests the correct display of emojis in headings at different levels. Particularly relevant is the encoding in PDF bookmarks and the table of contents.

## 🎯 Test scenarios

Emojis in headings place special demands on document processing:

- **PDF bookmarks**: Correct Unicode encoding in PDF table of contents
- **TOC generation**: Table of contents with emoji characters
- **Font fallbacks**: Switching between text and emoji fonts
- **Hierarchy**: Emojis at all heading levels (H1-H6)

## 📋 Emoji test

### Sample set

This page places emojis in headings to test bookmarks/TOC and PDF strings.

#### 🧪 Heading with emoji

Inline: ✅ ⚠️ ℹ️ 🔒 🔑 ♻️

#### 🧩 ZWJ sequences (complex)

👩‍💻 👨‍🔬 🧑‍🎨 👫

#### 🏁 Flags in text

🇩🇪 🇪🇺 🇬🇧 🇺🇸 🇺🇳

#### 🔢 Keycaps & variants

0 1 2 3 4 5 6 7 8 9 🔟 # *

# Template for multilingual neutral text

This template provides guidelines for creating content suitable for all language versions.

## Principles

Multilingual neutral content:

- **Cultural neutrality**: Avoid culture-specific references, idioms, or examples
- **Universal concepts**: Use internationally recognised ideas and terminology
- **Technical focus**: Emphasise technical accuracy over cultural context
- **Symbol preference**: Use symbols, diagrams, and code over prose where possible

## Language considerations

### Avoid

❌ **Culture-specific examples:**

```
Like preparing a traditional Sunday roast...
As American as apple pie...
```

❌ **Regional idioms:**

```
It's raining cats and dogs
The proof is in the pudding
```

❌ **Country-specific references:**

```
As required by UK GDPR...
Similar to the US ZIP code system...
```

### Prefer

✅ **Universal examples:**

```
Like preparing a meal...
A widely recognised pattern...
```

✅ **Clear, literal language:**

```
Heavy rainfall
Evidence demonstrates that...
```

✅ **International standards:**

```
As required by ISO 8601...
Following RFC 3339 date format...
```

## Content patterns

### Technical documentation

Technical content is naturally more neutral:

```
## Installation

1. Download the package
2. Extract to a directory
3. Run the installer
```

```
4. Verify installation with `command --version`
```

#### Code examples

Code transcends language barriers:

```
# Universal technical concepts
def calculate_total(items):
    return sum(item.price for item in items)
```

#### Mathematical notation

Mathematics is international:

```
The Pythagorean theorem: $a^2 + b^2 = c^2$
```

#### Visual elements

Diagrams and symbols work across languages:

- Flowcharts
- Sequence diagrams
- Icons and symbols (Unicode)
- Tables and matrices

### Metadata structure

For multilingual documents:

```
---
title: Your Title
date: YYYY-MM-DD
version: X.Y
doc_type: chapter  # or appropriate type
language_neutral: true  # Flag for neutral content
translation_notes: "Focus on technical accuracy"
---
```

### Testing checklist

Before publishing multilingual content:

- ☐ No culture-specific references
- ☐ No idioms or colloquialisms
- ☐ Technical terms properly defined
- ☐ Code examples are universal
- ☐ Numbers and dates use ISO formats
- ☐ Currency symbols avoided (use generic "units")
- ☐ Time zones specified if relevant
- ☐ Measurements use metric (SI) units

### Translation workflow

When translating neutral content:

1. **Preserve structure**: Keep headings and formatting identical

2. **Technical accuracy**: Verify technical terms in target language
3. **Literal translation**: Avoid creative interpretation
4. **Code unchanged**: Never translate code variable names or commands
5. **Metadata sync**: Keep version and date metadata consistent

# Templates

This directory contains reusable templates and patterns for documentation.

## Purpose

Templates provide:

- **Consistency**: Standardised structure across similar content
- **Efficiency**: Quick starting points for new documents
- **Quality**: Pre-validated formatting and metadata
- **Guidance**: Examples of best practices

## Available templates

### Multilingual neutral text

Template for content that must work across all language versions:

- Neutral cultural references
- Internationally recognised examples
- Language-independent code samples
- Universal symbols and notation

See multilingual-neutral-text.md for details.

## Template structure

Each template includes:

```yaml
---
title: Template Name
date: YYYY-MM-DD
version: X.Y
doc_type: template
show_in_summary: false  # Usually hidden from main TOC
---
```

## How to use templates

1. **Copy** the template file to your target location
2. **Rename** to match your content purpose
3. **Update** frontmatter (title, date, version, doc_type)
4. **Replace** template content with your material
5. **Validate** structure and formatting

## Template categories

### Content templates

- Chapter structures
- Example patterns
- Reference documentation layouts

**Metadata templates**

- Frontmatter configurations
- Navigation structures
- Build configurations

**Multilingual templates**

- Parallel translation frameworks
- Language-neutral content patterns
- Internationalisation guidelines

# Translator's Note

This document demonstrates multilingual publishing capabilities and translation workflows.

## Translation principles

When translating technical documentation:

- **Terminology consistency**: Maintain consistent translation of technical terms
- **Cultural adaptation**: Adapt examples and metaphors to target culture
- **Format preservation**: Keep structure, headings, and formatting identical
- **Technical accuracy**: Verify all code examples, commands, and references

## Language considerations

### British English conventions

This English version follows British English spelling and grammar conventions:

- Spelling: colour, organise, licence (noun)
- Punctuation: Single quotes for regular text, double for nested
- Date format: DD/MM/YYYY
- Number formatting: Comma for thousands (1,000)

### Unicode support

The document includes extensive Unicode content:

- **100+ languages**: Covering major writing systems
- **Emoji rendering**: Proper display of flags, symbols, and combined sequences
- **Right-to-left text**: Support for Arabic, Hebrew, and other RTL scripts

## Translation workflow

Content is maintained in parallel language directories:

```
de/     # German (Deutsch)
en/     # English (British)
```

Each language maintains:

- Independent SUMMARY.md (navigation structure)
- Language-specific metadata (book.json)
- Localised frontmatter and terminology

# List of Tables

This section provides a comprehensive index of all tables appearing throughout the document. Tables are numbered sequentially and referenced by their location in the text.

## Purpose

The list of tables serves multiple functions:

- **Quick reference**: Locate specific tables without scanning the entire document
- **Content overview**: Understand the range of comparative and structured information presented
- **Navigation aid**: Jump directly to tables of interest

## Organization

Tables are listed in order of appearance with:

- Table number
- Descriptive caption
- Page reference (in PDF output)

*Note: The complete list is automatically generated during the build process and includes all captioned tables from the chapters and appendices.*

# List of Figures

This section catalogues all figures, diagrams, and illustrations used throughout the document. Each figure is numbered and captioned for easy reference.

## Purpose

The list of figures provides:

- **Visual content index**: Overview of all graphical elements
- **Quick access**: Direct navigation to specific illustrations
- **Content audit**: Verification that all images are properly captioned

## Supported formats

The document includes figures in various formats:

- **Raster images**: PNG, JPEG for photographs and screenshots
- **Vector graphics**: SVG for scalable diagrams and icons
- **Mixed content**: Combination of different formats as needed

## Organization

Figures are listed sequentially with:

- Figure number
- Descriptive caption
- Page location (in PDF output)
- Format type where relevant

*Note: The complete list is automatically generated during the build process and includes all captioned figures from all document sections.*

# List of Abbreviations

This section defines abbreviations and acronyms used throughout the document.

## Technical abbreviations

**API**
Application Programming Interface

**CAP**
Consistency, Availability, Partition tolerance (theorem)

**CLI**
Command-Line Interface

**CPU**
Central Processing Unit

**CSS**
Cascading Style Sheets

**DPI**
Dots Per Inch

**HTML**
HyperText Markup Language

**HTTP**
HyperText Transfer Protocol

**IDE**
Integrated Development Environment

**ISO**
International Organization for Standardization

**JSON**
JavaScript Object Notation

**LTR**
Left-to-Right (text direction)

**PDF**
Portable Document Format

**PNG**
Portable Network Graphics

**RTL**
Right-to-Left (text direction)

**SQL**
Structured Query Language

**SVG**
Scalable Vector Graphics

**TOC**
Table of Contents

**UI**
User Interface

**URL**
Uniform Resource Locator

**UTF**
Unicode Transformation Format

**XML**
Extensible Markup Language

**YAML**
YAML Ain’t Markup Language

**ZWJ**
Zero Width Joiner (Unicode)

# Appendices

Supplementary materials, technical specifications, and reference information.

## Purpose

Appendices provide:

- **Supplementary detail**: In-depth technical information
- **Reference material**: Tables, specifications, and data
- **Technical documentation**: Implementation details and configurations
- **Supporting evidence**: Font coverage, testing results, methodologies

## Organisation

Appendices are labelled alphabetically:

- **Appendix A**: Data sources and table layout
- **Appendix B**: Emoji and font coverage

Each appendix includes:

- Unique identifier (A, B, C…)
- Descriptive title
- Category classification (technical, reference, etc.)
- Version history

## Structure

### Frontmatter

Each appendix uses consistent metadata:

```yaml
---
title: Appendix X — Title
date: YYYY-MM-DD
version: X.Y
doc_type: appendix
appendix_id: "X"
category: "technical" | "reference" | "legal"
---
```

### Content patterns

Appendices typically include:

- Technical specifications
- Data tables and matrices
- Testing methodologies
- Configuration examples
- Detailed calculations
- Reference implementations

## Navigation

Appendices appear:

- After main content chapters
- Before indices (table of contents, figures, etc.)
- In alphabetical order by identifier

They are accessible via:

- Table of contents links
- PDF bookmarks
- Cross-references from main text

## Cross-referencing

Reference appendices from main text:

```
See [Appendix A](../appendices/appendix-a.md) for data sources.
Font coverage is detailed in [Appendix B](../appendices/emoji-font-coverage.md).
```

## Types of appendices

### Technical appendices

- Implementation details
- Algorithm specifications
- Configuration references
- Testing procedures

### Reference appendices

- Data tables
- Glossaries
- Bibliography
- Standards references

### Legal appendices

- Licence texts
- Compliance documentation
- Attribution details
- Legal notices

# Appendix A – Data sources and table layout

This appendix documents the data sources and structural conventions used in tables throughout this document.

## Table design principles

### Readability

Tables are designed for:

- **Quick scanning**: Clear headers and consistent alignment
- **Data comparison**: Parallel structure for easy comparison
- **Reference use**: Complete information without requiring external context

### Consistency

All tables follow:

- Consistent column ordering
- Uniform header formatting
- Standard alignment rules (left for text, right for numbers)
- Descriptive captions

## Table types

### Comparative tables

Structure for comparing options:

| Feature | Option A | Option B | Option C |
|---|---|---|---|
| Performance | High | Medium | Low |
| Complexity | Low | Medium | High |
| Cost | Low | Medium | High |

### Reference tables

Data lookup format:

| Key | Value | Description |
|---|---|---|
| Term 1 | Definition | Detailed explanation |
| Term 2 | Definition | Detailed explanation |

### Multi-level tables

Hierarchical information:

| Category | Subcategory | Details |
|---|---|---|
| Type A | Variant 1 | Specifications |
| | Variant 2 | Specifications |
| Type B | Variant 1 | Specifications |

## Data sources

### Primary sources

Tables are compiled from:

- Official documentation and specifications
- Published standards (ISO, RFC, etc.)
- Peer-reviewed research where applicable
- Vendor documentation and release notes

### Data verification

All tabulated data:

1. Cross-referenced with primary sources
2. Verified for current accuracy
3. Dated to indicate currency
4. Linked to source documentation where possible

### Update policy

Tables are reviewed:

- During major version updates
- When underlying specifications change
- Following significant technology releases
- As corrections are identified

## Formatting conventions

### Numerical data

- **Integers**: No decimal separator (1000, not 1,000)
- **Decimals**: Period as decimal separator (3.14)
- **Percentages**: Number followed by % symbol (85%)
- **Ranges**: En dash between values (10–20)

### Text alignment

- **Left-aligned**: Text, descriptions, category names
- **Right-aligned**: Numbers, dates, versions
- **Centre-aligned**: Yes/No, checkmarks, symbols

### Special symbols

- ✓ = Supported/Yes
- ✗ = Not supported/No
- — = Not applicable
- ≈ = Approximately
- ≥/≤ = Greater/less than or equal

## Caption format

Table captions include:

```
Table X.Y: Descriptive title
```

Where:

- X = Chapter number
- Y = Sequential table number within chapter
- Title describes content succinctly

## Accessibility

### Screen readers

Tables use:

- Proper Markdown table syntax for correct HTML rendering
- Descriptive headers that work when read sequentially
- Captions that provide context independent of surrounding text

### Print readability

Table design considers:

- Page width constraints in PDF output
- Readability at standard print sizes
- Clear distinction between header and data rows

### Example table

| Item | Purpose |
|---|---|
| Heading | TOC/bookmarks |
| Table | list of tables |

### Example code block

```
python -m gitbook_worker.tools.workflow_orchestrator --help
```

# Appendix B – Emoji & font coverage

This appendix documents font coverage for the diverse Unicode content used throughout this document, including emoji rendering and multilingual text support.

## Font stack

The document uses a carefully configured font stack:

### Primary text fonts

#### DejaVu Serif / DejaVu Sans

- **Coverage**: Latin, Cyrillic, Greek, basic IPA
- **Purpose**: Main body text and headings
- **Licence**: Free (Bitstream Vera derivative)
- **Unicode blocks**: ~3,000 glyphs covering common scripts

### Emoji fonts

#### Twemoji Mozilla (COLRv1)

- **Coverage**: Full Emoji 13.0+ support
- **Format**: COLRv1 (colour font format)
- **Purpose**: Primary emoji rendering
- **Licence**: CC BY 4.0
- **Rendering**: Native colour in modern systems

#### Twitter Color Emoji (Fallback)

- **Coverage**: Emoji 12.0
- **Format**: CBDT/CBLC (bitmap colour)
- **Purpose**: Fallback for older systems
- **Licence**: CC BY 4.0 / MIT

## Emoji categories tested

Comprehensive testing across all Unicode emoji categories:

### 😀 People & Emotions

- Faces: 😀 😃 😄 😁 😅
- Hands: 👋 🤚 🖐 ✋ 🖖
- People: 👶 👧 🧒 👦 👧
- Skin tones: 👋🏻 👋🏼 👋🏽 👋🏾 👋🏿

### 🐐 Animals & Nature

- Mammals: 🐐 🐏 🐄 🐄 🐖
- Birds: 🐔 🐓 🐤 🐣 🐥
- Plants: 🌲 🌳 🌴 🌵 🌾
- Weather: ☀️ ⛅ ☁️ ⛈ 🌧

### 🍕 Food & Drink

- Prepared food: 🍕 🍔 🍟 🌭 🥪
- Fruit: 🍎 🍊 🍋 🍌 🍉
- Drinks: 🍵 🍏 🥤 🍺 🍷

### ⚽ Activities & Sports

- Sports: ⚽ 🏀 🏈 🎾 🎾
- Games: 🎮 🎯 🎲 🎰 🎳
- Arts: 🎨 🎭 🎪 🎬 🎤

### 🚗 Travel & Places

- Vehicles: 🚗 🚕 🚙 🚌 🚎
- Buildings: 🏠 🏡 🏢 🏥 🏪
- Geography: 🏔 🏕 🏜 🏝 🌵

### 💡 Objects

- Tech: 💻 ⌨ 🖥 🖨 🖱
- Tools: 🔨 🔨 🛠 🛠 🔧
- Office: 📝 🔪 🔪 🖌 🖌

### 🔣 Symbols

- Math: ➕ ➖ ✖ ➗ ➖
- Arrows: ⬆ ⬇ ⬅ ➡ ↔
- Shapes: ⬛ ⬜ 🔲 ⬜ ⬛

### 🏁 Flags

- Country flags: 🇬🇧 🇩🇪 🇫🇷 🇪🇸 🇮🇹
- Regional flags: 🏴 (requires ZWJ support)
- Special flags: 🏳 🏴 🇲🇳

## Complex emoji sequences

### Zero-Width Joiner (ZWJ) sequences

Testing compound emoji:

- **Family**: 👨‍👩‍👧‍👦 (requires ZWJ support)
- **Professions**: 👨‍👩‍👧‍👦 🧑‍🏫 🧑‍🏫
- **Combinations**: 🏴 🇲🇳

### Skin tone modifiers

Fitzpatrick scale support:

- Type 1-2 (light): 👋
- Type 3 (medium-light): 👋
- Type 4 (medium): 👋
- Type 5 (medium-dark): 👋
- Type 6 (dark): 👋

### Flag sequences

Regional indicator symbols:

- 🇬 + 🇧 = 🇬🇧 (UK flag)
- 🇩 + 🇪 = 🇩🇪 (German flag)

## Script coverage

Multilingual text support across 100+ languages:

### Latin-based scripts

- Western European: English, German, French, Spanish
- Eastern European: Polish, Czech, Hungarian
- Special characters: Ā Ē Ī Ō Ū (macrons)

### Cyrillic

- Russian: Привет мир
- Ukrainian: Привіт світ
- Bulgarian: Здравей свят

### Greek

- Modern Greek: Γεια σου κόσμε
- Polytonic Greek: ἀρχή (archaic)

### Asian scripts

- Chinese (Simplified): 你好世界
- Japanese: こんにちは世界 (Hiragana)
- Korean: 안녕하세요 세계 (Hangul)

### Arabic & RTL scripts

- Arabic: مرحبا بالعالم (RTL)
- Hebrew: שלום עולם (RTL)
- Persian: سلام دنیا (RTL)

### South Asian scripts

- Devanagari: नमस्ते दुनिया (Hindi)
- Tamil: வணக்கம் உலகம்
- Bengali: নমস্কার বিশ্ব

### Other scripts

- Thai: สวัสดีชาวโลก
- Amharic: ሰላም ዓለም
- Georgian: გამარჯობა მსოფლიო

## Testing methodology

### Visual verification

All emoji and scripts:

1. Rendered in PDF output
2. Visually inspected for correctness
3. Checked for proper colour rendering (emoji)
4. Verified in both screen and print modes

### Font fallback chain

The system tests fallback behaviour:

`Primary → Secondary → System fallback`

- If primary font lacks a glyph, system tries secondary
- Final fallback to system fonts if needed
- Missing glyphs indicated by □ (replacement character)

### Known limitations

1. **ZWJ sequences**: Complex emoji may render as separate glyphs on older systems
2. **COLRv1 support**: Requires modern font rendering (Cairo 1.18+, FreeType 2.13+)
3. **RTL layout**: Simplified handling; complex bidirectional text may need adjustment
4. **Rare scripts**: Some scripts require additional font installation

## Font configuration

See `fonts-storage/fonts.conf` for the complete fontconfig configuration.

Key settings:

- Emoji font priority ordering
- Script-specific font mappings
- Fallback chains
- Hinting and antialiasing preferences- YAML frontmatter (document metadata)
- Heading hierarchy (TOC / PDF bookmarks)
- Lists, code blocks, blockquotes
- Tables and references
- Stable navigation (SUMMARY.md)

### Example table

| Item | Purpose |
|---|---|
| Heading | TOC/bookmarks |
| Table | list of tables |

### Example code block

```
python -m gitbook_worker.tools.workflow_orchestrator --help
```

# Legal Notice

This document serves as a demonstration of legal notice formatting in technical publications.

## Publisher information

In a production document, this section would include:

- Publisher name and address
- Responsible parties
- Editorial team contact information
- ISBN/ISSN numbers where applicable

## Copyright notice

Typical copyright statements include:

- Copyright year and holder
- Rights reserved statement
- Permitted use conditions
- Trademark acknowledgements

## Licence terms

For open-source documentation:

- **Content licence**: Creative Commons or similar
- **Code licence**: MIT, Apache, GPL, or other open-source licence
- **Asset licences**: Individual licences for fonts, images, and third-party content

See LICENSE-CODE and LICENSE-FONTS for specific terms.

## Liability disclaimer

Standard disclaimers typically cover:

- Accuracy of information
- Fitness for particular purpose
- Third-party content responsibility
- External link liability

## Data protection

For digital publications:

- Data collection practices
- Privacy policy references
- Cookie usage (web versions)
- Analytics and tracking disclosure

## Contact

In production, include:

- Technical support contact
- Editorial feedback address

- Legal enquiries contact

# Glossary

Definitions of technical terms used throughout this document.

## A

**API** (Application Programming Interface)
Interface that enables software components to communicate with each other.

**Accessibility**
Design of content that is usable by people with disabilities.

## B

**Bibliography**
List of sources cited or referenced in a document.

**Build Pipeline**
Automated process for converting source files into output formats.

## C

**CI/CD** (Continuous Integration / Continuous Deployment)
Practice of frequently integrating code and automatically deploying it.

**COLRv1**
Modern colour font format for vector graphics in fonts.

## D

**Documentation Framework**
Structured system for creating and managing documentation.

## E

**Emoji**
Pictographic characters from the Unicode Standard representing emotions and objects.

## F

**Fontconfig**
Library for configuring and customising font access.

**Frontmatter**
Metadata block at the beginning of a Markdown file (YAML format).

## G

**Git**
Distributed version control system for tracking code changes.

**Glyph**
Visual character representing one or more Unicode code points.

## I

**ISO 8601**
International standard for date and time formats.

## L

**LaTeX**
Typesetting system for high-quality typographic output.

**Licence**
Legal agreement regarding the use of software or content.

## M

**Markdown**
Lightweight markup language for formatting text.

**Metadata**
Information about documents (title, author, date, etc.).

## O

**Open Source**
Software with freely available source code.

**OpenType**
Modern font format with advanced typographic capabilities.

## P

**Pandoc**
Universal document conversion tool.

**PDF** (Portable Document Format)
Platform-independent file format for documents.

## R

**Rendering**
Process of visually displaying code or markup.

**RTL** (Right-to-Left)
Text direction from right to left (Arabic, Hebrew).

## S

**Semantic Versioning**
Version numbering using the MAJOR.MINOR.PATCH scheme.

**SVG** (Scalable Vector Graphics)
Vector graphics format for scalable images.

## U

**Unicode**
Universal character encoding standard for all writing systems.

## V

**Version Control**
System for tracking and managing changes to files.

## X

**XeLaTeX**
LaTeX engine with native Unicode and OpenType support.

## Y

**YAML** (YAML Ain't Markup Language)
Human-readable data serialisation format.

## Z

**ZWJ** (Zero Width Joiner)
Invisible Unicode character for combining emojis.

---

*Note: This glossary contains terms relevant to this documentation framework. For complete definitions, please consult official specifications and standards.*

# Citations & further reading

Bibliography and additional resources for further reading.

## Purpose

This bibliography:

- **Documents sources**: All cited references
- **Enables verification**: Readers can check original sources
- **Provides context**: Background information on topics
- **Extends knowledge**: Further reading materials

## Citation style

This document uses **APA style** (7th edition):

```
Author, A. A. (Year). Title of work. Publisher.
```

For online resources:

```
Author, A. A. (Year). Title. Website Name. URL
```

## Categories

### Technical standards

Official specifications and standards:

- ISO, RFC, W3C specifications
- Unicode Consortium documents
- OpenType specifications

### Documentation

Official tool and software documentation:

- Pandoc manual
- LaTeX/XeLaTeX references
- Git documentation
- Python libraries

### Articles and tutorials

Best practices and guides:

- Technical blog posts
- Tutorial websites
- Community resources

### Books

Technical books on relevant topics:

- Documentation methodology
- Typography and typesetting
- Software development

## Example entries

### Standards

**Unicode Consortium.** (2023). *The Unicode Standard, Version 15.0*. Unicode Consortium. https://www.unicode.org/versions/Unicode15.0.0/

**Internet Engineering Task Force.** (2018). *RFC 8259: The JavaScript Object Notation (JSON) Data Interchange Format*. IETF. https://tools.ietf.org/html/rfc8259

### Software documentation

**Pandoc.** (2023). *Pandoc User's Guide*. https://pandoc.org/MANUAL.html

**LaTeX Project.** (2023). *LaTeX2e: An unofficial reference manual*. https://latexref.xyz/

### Articles

**Semantic Versioning.** (2023). *Semantic Versioning 2.0.0*. https://semver.org/

**Markdown Guide.** (2023). *Basic Syntax*. https://www.markdownguide.org/basic-syntax/

## Further resources

### Online communities

- **Stack Overflow**: Questions and answers on technical problems
- **GitHub**: Open-source projects and discussions
- **Reddit**: r/LaTeX, r/Markdown, r/technicalwriting

### Learning platforms

- **Write the Docs**: Community for technical writers
- **Overleaf**: Online LaTeX editor with tutorials
- **GitHub Learning Lab**: Git and GitHub courses

### Tools

- **Zotero**: Reference management
- **Grammarly**: Language checking
- **draw.io**: Diagram creation

## Source verification

When using sources:

1. **Check currency**: Is the information still current?
2. **Assess authority**: Is the source trustworthy?
3. **Multiple sources**: Confirm information
4. **Primary sources**: Prefer official documentation

## Contribution guidelines

When adding new references:

- Consistent citation style (APA)
- Complete bibliographic information

- Access date for online resources
- Categorisation for easy navigation

---

*Note: This bibliography is continuously updated. Contributions and corrections are welcome.*

# Index

Alphabetical subject index for quick access to topics.

## Purpose

The index enables:

- **Quick lookup**: Immediate access to specific terms
- **Cross-references**: Linking related concepts
- **Completeness**: Overview of covered topics
- **Navigation**: Alternative access pattern to table of contents

## Structure

The index is organised:

- **Alphabetically**: Sorted by initial letter
- **Hierarchically**: Main and sub-terms
- **With page references**: Direct links to sections
- **Cross-referenced**: "See also" notes

## Usage

### In printed versions

The index appears:

- At the end of the document
- After appendices and lists
- With page numbers for each reference

### In digital versions

The index provides:

- Clickable links to sections
- Search functionality within the index
- Integration with PDF bookmarks

## Indexing

### Entries

Typical index entries:

```
Term, Page
  Sub-term, Page
  Sub-term, Page
Another Term, Page
  see also: Related Term
```

### Conventions

- **Bold**: Primary definition or main discussion
- *Italic*: Passing mention

- (Figure): Visual representation
- (Table): Tabular information

## Automatic generation

This index can be automatically generated from:

- Explicit index markers in Markdown
- Headings and subsections
- Glossary entries
- Code example titles

## Best practices

For effective indexing:

1. **Consistent terms**: Use uniform terminology
2. **Multiple entries**: Index concepts under different search terms
3. **Cross-references**: Connect related terms
4. **Avoid over-indexing**: Include only significant references

## Maintenance

The index should be:

- Updated with each major version
- Include new terms from added chapters
- Remove obsolete references
- Check consistency with glossary

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*Note: A complete index is generated during the final build process and includes all indexed terms with precise page references.*

# Acknowledgments & Attributions

This document acknowledges the contributors, tools, and resources that made this publication possible.

## Font attributions

This document uses the following open-source fonts:

### Twemoji Mozilla

- **Licence**: CC BY 4.0
- **Source**: Mozilla's Twemoji COLRv1 implementation
- **Purpose**: Emoji rendering in text
- **Licence URL**: https://creativecommons.org/licenses/by/4.0/

### DejaVu Fonts

- **Licence**: Bitstream Vera Licence / Arev Licence
- **Purpose**: Base text rendering
- **Coverage**: Latin, Cyrillic, Greek, and extensive Unicode blocks

### Twitter Color Emoji

- **Licence**: CC BY 4.0 (artwork) / MIT (code)
- **Source**: Twitter's open-source emoji set
- **Purpose**: Fallback emoji rendering

## Software tools

Built with open-source software:

- **Python**: Core automation and orchestration
- **Pandoc**: Markdown to LaTeX conversion
- **XeLaTeX/LuaLaTeX**: PDF typesetting
- **GitBook**: Content structure and metadata

## Python libraries

Key dependencies:

- **PyYAML**: Configuration and frontmatter parsing
- **GitPython**: Git repository management
- **Jinja2**: Template processing
- **svglib**: SVG handling and conversion

## Content and methodology

Special acknowledgements:

- **Unicode Consortium**: For comprehensive character encoding standards
- **OpenType specification**: For modern font rendering capabilities
- **Markdown community**: For lightweight, readable markup language

## Contributors

Gratitude to all who contributed:

- Content authors and editors
- Technical reviewers
- Translation teams
- Testing and quality assurance
- Documentation framework developers

## Licence compliance

All third-party assets are used in accordance with their respective licences. See:

- LICENSE-CODE for code licencing
- LICENSE-FONTS for font licencing
- Individual attribution files in `fonts-storage/` for detailed font information

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*This acknowledgements section demonstrates proper attribution practices for open-source documentation projects.*

# Errata

This section documents corrections and updates to the published document.

## Purpose

The errata page serves to:

- Document errors discovered after publication
- Provide corrections for known issues
- Track version-specific changes
- Maintain document accuracy over time

## How to report issues

If you discover an error:

1. Check this page to see if it's already documented
2. Note the version number, page/section, and nature of the issue
3. Report via the appropriate channel (issue tracker, email, etc.)

## Errata format

Each entry includes:

- **Version**: Which version contains the error
- **Location**: Page number or section reference
- **Type**: Typographical, technical, factual, or formatting error
- **Description**: What is incorrect
- **Correction**: The correct information
- **Status**: Fixed in version X.X.X or pending

## Version 1.0.0

*No errata reported for this version.*

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## Continuous improvement

This document is maintained as a living record. Regular reviews ensure:

- Technical accuracy
- Up-to-date references
- Correction of typographical errors
- Improvement of clarity

Check the release notes for the current version status.

# Release Notes

This document tracks changes, improvements, and fixes across versions.

## Version 1.0.0 (2024-06-01)

### Initial release

First public version of the documentation framework.

**Features:**

- Multilingual support (English and German)
- Comprehensive emoji rendering across all Unicode categories
- 100+ language samples demonstrating font coverage
- Professional PDF generation with proper typography
- Structured navigation with table of contents
- Code examples and technical documentation patterns

**Content structure:**

- Core chapters demonstrating documentation patterns
- Examples section (emoji tests, image formats, language samples)
- Appendices (technical specifications, font coverage)
- Complete metadata framework (YAML frontmatter)

**Technical foundation:**

- Python-based build orchestration
- Markdown source format
- LaTeX/XeLaTeX PDF generation
- Unicode and OpenType font support
- Automated table of contents generation

### Known limitations

- Some complex emoji sequences may render differently depending on font support
- RTL (right-to-left) text layout uses simplified handling
- Large SVG images may require optimization for faster rendering

### Requirements

- Python 3.8+
- XeLaTeX or LuaLaTeX
- Required fonts: DejaVu, Twemoji Mozilla
- Git for version control

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## Version history format

Future releases will follow this structure:

### Version X.Y.Z (YYYY-MM-DD)

**Added:**

- New features and capabilities

**Changed:**

- Modifications to existing functionality

**Fixed:**

- Bug fixes and corrections

**Deprecated:**

- Features marked for future removal

**Removed:**

- Discontinued features

**Security:**

- Security-related changes

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## Semantic versioning

This project follows Semantic Versioning:

- **MAJOR** (X.0.0): Incompatible changes
- **MINOR** (0.X.0): Backwards-compatible new features
- **PATCH** (0.0.X): Backwards-compatible bug fixes

# Colophon

This document was created using a modern publishing workflow that transforms Markdown source files into professional PDF output.

## Production details

### Typography

- **Body text**: Professional serif typeface
- **Headings**: Sans-serif for clear hierarchy
- **Code**: Monospace font for technical content
- **Emoji**: Colour emoji font with extensive Unicode coverage

### Software and tools

This document was produced using:

- **Python**: Workflow orchestration and document processing
- **Markdown**: Lightweight markup for source content
- **LaTeX**: Professional typesetting engine
- **Git**: Version control for source management

### Document format

- **PDF/A compliance**: Archival-quality output
- **Embedded fonts**: Complete font embedding for consistency
- **Bookmarks**: Hierarchical navigation structure
- **Metadata**: Comprehensive document properties

### Design principles

The visual design follows established principles:

- Clear typographic hierarchy
- Generous whitespace for readability
- Consistent formatting throughout
- Accessible colour contrasts

### Licensing

See the licensing sections for details on content, code, and font licenses.

### Revision

Version 1.0, January 2026